

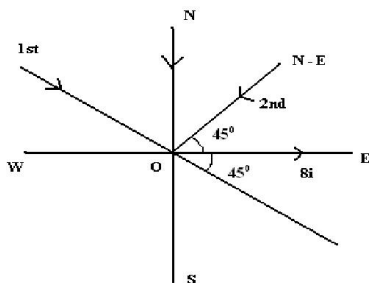
LEVEL-VI

SINGLE ANSWER QUESTIONS

1. The unit vector $\vec{c}$ if $-\hat{i} + \hat{j} - \hat{k}$ bisects the angle between vectors $\vec{c}$ and $3\hat{i} + 4\hat{j}$ is

- (A) $-\frac{1}{15}(11\hat{i} + 10\hat{j} - 2\hat{k})$
 (B) $\frac{1}{15}(-11\hat{i} - 10\hat{j} - 2\hat{k})$
 (C) $\frac{1}{15}(11\hat{i} - 10\hat{j} - 2\hat{k})$
 (D) $\frac{1}{15}(11\hat{i} - 10\hat{j} + 2\hat{k})$

2. A man travelling east at 8km/h finds that the wind seems to blow directly from the north. On doubling the speed, he finds that it appears to come from the north-east. The velocity of the wind is



- (A) $\theta = 45^\circ, v = 8\sqrt{2}$ (B) $\theta = -45^\circ, v = 8\sqrt{2}$
 (C) $\theta = 90^\circ, v = 4\sqrt{2}$ (D) $\theta = 60^\circ, v = 8\sqrt{2}$

3. ABCD is a parallelogram. If L and M be the middle points of BC and CD, respectively express $\vec{AL}$ and $\vec{AM}$ in terms of $\vec{AB}$ and $\vec{AD}$. If $\vec{AL} + \vec{AM} = K \cdot \vec{AC}$ then the value of K is

- (A) $1/2$ (B) $3/2$ (C) 2 (D) 1

4. If the vectors $\vec{a}$ and $\vec{b}$ are linearly independent satisfying

$$(\sqrt{3} \tan \theta + 1)\vec{a} + (\sqrt{3} \sec \theta - 2)\vec{b} = 0, \text{ then the most general values of } \theta \text{ are}$$

VECTOR ALGEBRA

(A) $n\pi - \frac{\pi}{6}, n \in \mathbb{Z}$ (B) $2n\pi \pm \frac{11\pi}{6}, n \in \mathbb{Z}$

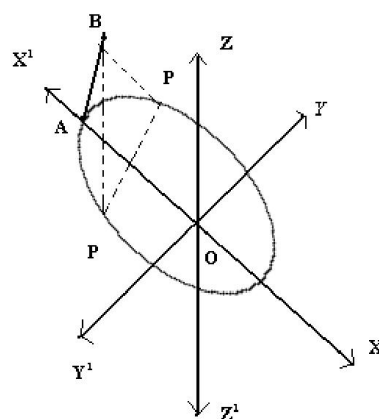
(C) $2n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$ (D) $2n\pi \pm \frac{11\pi}{6}, n \in \mathbb{Z}$

5. If $\vec{b}$ is vector whose initial point divides the join of $5\hat{i}$ and $5\hat{j}$ in the ratio $k:1$ and whose terminal point is the origin and $|\vec{b}| \leq \sqrt{37}$, then k lies in the interval

(A) $\left[-6, \frac{-1}{6}\right]$ (B) $(-\infty, -6] \cup \left[\frac{-1}{6}, \infty\right)$

(C) $[0, 6]$ (D) $\left(-6, \frac{-1}{5}\right]$

6. Let $x^2 + 3y^2 = 3$ be the equation of an ellipse in the xy-plane. A and B are two points whose position vectors are $-\sqrt{3}\hat{i}$ and $-\sqrt{3}\hat{i} + 2\hat{k}$ then the position vector of a point P on the ellipse such that $\angle APB = \pi/4$ is



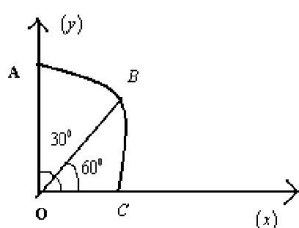
(A) $\pm \hat{j}$ (B) $\pm(\hat{i} + \hat{j})$

(C) $\pm \hat{i}$ (D) $\pm \vec{k}$

7. If vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$, $\vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$ constitute the sides of $\triangle ABC$, then the length of the median bisecting the vector $\vec{c}$

- (A) $\sqrt{2}$ (B) $\frac{\sqrt{14}}{2}$ (C) $\sqrt{74}$ (D) $\sqrt{6}$

8. Let AC be an arc of a circle, subtending a right angle at the centre O. The point B divides the arc AC in the ratio 1:2. If $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$, then calculate $\vec{OC}$ in terms of $\vec{a}$ and $\vec{b}$



- (A) $2\vec{b} + \sqrt{3}\vec{a}$ (B) $2\vec{b} - \sqrt{3}\vec{a}$
 (C) $-2\vec{b} - \sqrt{3}\vec{a}$ (D) $3\vec{b} + \sqrt{2}\vec{a}$
9. ABC is a triangle and O any point in the same plane. AO, BO and CO meet the sides BC, CA and AB respectively at points D, E and F then $\left| \frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF} \right|$
 (A) 1 (B) 1/2 (C) 1/3 (D) 2
10. In triangle ABC, $\angle A = 30^\circ$, H is the orthocenter and D is the mid-point of BC. Segment HD is produced to T such that $HD = DT$. The length AT is equal to
 A) 2BC B) 3BC C) $\frac{4}{3}BC$ D) $\frac{1}{3}BC$
11. The value of λ so that the points P, Q, R and S on the sides OA, OB, OC and AB of a regular tetrahedron OABC are coplanar when $\frac{OP}{OA} = \frac{1}{3}, \frac{OQ}{OB} = \frac{1}{2}, \frac{OR}{OC} = \frac{1}{3}$ and $\frac{OS}{AB} = \lambda$ then
 A) $\lambda = \frac{1}{2}$ B) $\lambda = -1$
 C) $\lambda = 0$ D) for no value of λ
12. In triangle ABC, AD and AD' are the bisectors of the angle A meeting BC in D and D' respectively. A' is the mid point of DD'; B' and C' are the points on CA and AB similarly obtained then A', B', C' forms

- A) Equilateral triangle B) Isosceles triangle
 C) Scalene triangle D) Straight line

13. Any plane cuts the sides AB, BC, CD, DA of a quadrilateral in P, Q, R, S respectively,

and if $\frac{AP}{PB} = \lambda_1, \frac{BQ}{QC} = \lambda_2, \frac{CR}{RD} = \lambda_3,$

$\frac{DS}{SA} = \lambda_4$. Then $\lambda_1 \lambda_2 \lambda_3 \lambda_4 =$

- A) 1 B) 2 C) 3 D) 4

14. Points X and Y are taken on the sides QR and RS respectively of a parallelogram PQRS. So that $QX = 4XR$ and $RY = 4YS$ the

line XY cuts the line PR at Z then $\frac{PZ}{ZR} =$

- A) $\frac{1}{4}$ B) $\frac{21}{5}$ C) $\frac{21}{4}$ D) $\frac{16}{3}$

15. Let OABCD is a pentagon in which the side OA and CB are parallel and the sides OD and AB are parallel and $OA : CB = 2 : 1$ and $OD : AB = 1 : 3$. If the diagonal OC and AD meet at X then $\frac{OX}{XC} =$

- A) $\frac{1}{3}$ B) $\frac{1}{5}$ C) $\frac{2}{3}$ D) $\frac{2}{5}$

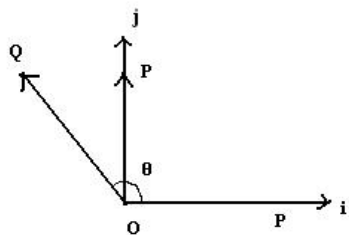
16. Let $\vec{a} = \vec{i} + \vec{j}, \vec{b} = \vec{j} - \vec{k}$. If 'm' be slope of tangent to the curve $y = f(x)$ at $x = -1$

where $f\left(\frac{3x-4}{3x+4}\right) = x+2 \quad \forall x \in \mathbb{R} - \left\{-\frac{4}{3}\right\}$

then the vector of magnitude "m" along the bisector of angle between $\vec{a}$ and $\vec{b}$ is

- A) $\frac{3\sqrt{3}}{\sqrt{2}}(\vec{i} + 2\vec{j} - \vec{k})$ B) $\frac{\sqrt{2}}{\sqrt{3}}(\vec{i} - \vec{j} + 2\vec{k})$
 C) $\frac{\sqrt{2}}{3\sqrt{3}}(\vec{i} + 2\vec{j} - \vec{k})$ D) $\frac{\vec{i} + \vec{j} - \vec{k}}{\sqrt{2}}$

17. If the resultant of two forces is equal in magnitude to one of the components and perpendicular to it in direction, find the other component using the vector method



- (A) 135° (B) 120° (C) 90° (D) 45°

18. If the vector $\vec{b} = \left(\tan \alpha, -1, 2\sqrt{\sin \frac{\alpha}{2}} \right)$ and

$$\vec{c} = \left(\tan \alpha, \tan \alpha, -\frac{3}{\sqrt{\sin \frac{\alpha}{2}}} \right) \text{ are orthogonal}$$

and a vector $\vec{a} = (1, 3, \sin 2\alpha)$ makes an obtuse angle with the z-axis then the value of α is

- A) $\alpha = (4n+1)\pi - \tan^{-1} 2$
 B) $\alpha = (4n+2)\pi - \tan^{-1} 2$
 C) $\alpha = (4n+1)\pi + \tan^{-1} 2$
 D) $\alpha = (4n+2)\pi + \tan^{-1} 2$

19. Image of the point P with position vector $7\hat{i} - \hat{j} + 2\hat{k}$ in the line whose vector equation is $\vec{r} = 9\hat{i} + 5\hat{j} + 5\hat{k} + \lambda(\hat{i} + 3\hat{j} + 5\hat{k})$ has the position vector

- A) $-9\hat{i} + 5\hat{j} + 2\hat{k}$ B) $9\hat{i} + 5\hat{j} - 2\hat{k}$
 C) $9\hat{i} - 5\hat{j} - 2\hat{k}$ D) $9\hat{i} + 5\hat{j} + 2\hat{k}$

20. Foot of the perpendicular from the point $P(\vec{a})$ to the line $\vec{r} = \vec{b} + t\vec{c}$

A) $\vec{b} + \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c}$ B) $\vec{b} - \left(\frac{\vec{a} \cdot \vec{c}}{|\vec{c}|^2} \right) \vec{c}$

C) $\vec{a} + \left(\frac{(\vec{b} - \vec{c}) \cdot \vec{c}}{|\vec{c}|^2} \right) \vec{c}$ D) None

21. The reflection of the point $P(\vec{a})$ with respect to the line $\vec{r} = \vec{b} + t\vec{c}$

A) $2\vec{b} + \vec{a} + 2 \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c}$

B) $\vec{b} + \vec{a} + \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c}$

C) $\vec{b} + \vec{a} + \left(\frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \right) \vec{a}$ D) None

22. The $\perp^{\text{lar}}$ distance from $A(1, 4, -2)$ from the segment BC where $B = (2, 1, -2)$, $C(0, -5, 1)$

A) $\frac{3\sqrt{26}}{7}$ B) $\frac{\sqrt{26}}{7}$ C) $\frac{3}{7}$ D) $\sqrt{26}$

23. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that

$$|\vec{a} + 2\vec{b} + 3\vec{c}| = \sqrt{3 + 2\sqrt{2}}, \text{ angle between } \vec{a}$$

and $\vec{b}$ is α , angle between $\vec{a}$ and $\vec{c}$ is β and angle between $\vec{b}$ and $\vec{c}$ varies in

$$\left[\frac{\pi}{2}, \frac{2\pi}{3} \right] \text{ then the greatest value of}$$

$$4 \cos \alpha + 6 \cos \beta$$

- A) $2\sqrt{2} + 5$ B) $-2\sqrt{2} + 5$
 C) $2\sqrt{2} - 5$ D) $\sqrt{42}$

24. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}, 3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral PQRS must be a [IIT JEE 2010]

- (A) parallelogram, which is neither a rhombus nor a rectangle
 (B) square
 (C) rectangle, but not a square
 (D) rhombus, but not a square

25. Two adjacent sides of a parallelogram ABCD are given by $\vec{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\vec{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an acute angle α in the plane of the

parallelogram so that AD becomes AD^1 . If AD^1 makes a right angle with the side AB, then the cosine of the angle α is given by [IIT JEE 2010]

- (A) $\frac{8}{9}$ (B) $\frac{\sqrt{17}}{9}$ (C) $\frac{1}{9}$ (D) $\frac{4\sqrt{5}}{9}$

26. The vector $\vec{A}$ satisfying the vector equation $\vec{A} + \vec{B} = \vec{a}$, $\vec{A} \times \vec{B} = \vec{b}$ and $\vec{A} \cdot \vec{a} = 1$, where $\vec{a}$ and $\vec{b}$ are given vectors is

A) $\frac{(\vec{b} \times \vec{a}) + \vec{a}(a^2 - 1)}{a^2}$ B) $\frac{(\vec{a} \times \vec{b}) + \vec{a}}{a^2}$

C) $\frac{\vec{a}(a^2 - 1) + \vec{b}(b^2 - 1)}{a^2 + b^2}$

D) None of these

27. The point of intersection of the lines $\vec{r} \times \vec{a} = \vec{b} \times \vec{a}$ and $\vec{r} \times \vec{b} = \vec{a} \times \vec{b}$

- A) $\vec{a} - \vec{b}$ B) $\vec{a} + \vec{b}$
C) $2\vec{a} + 3\vec{b}$ D) $3\vec{a} - 2\vec{b}$

28. In a regular tetrahedron of side a the distance of any vertex from opposite face is

- A) $\sqrt{\frac{2}{3}}a$ B) $\sqrt{\frac{1}{3}}a$ C) $\frac{\sqrt{2}}{3}a$ D) $\frac{1}{\sqrt{2}}a$

29. the shortest distance between two opposite edges regular tetrahedron of side k is

- A) $\frac{k}{2}$ B) $\frac{k}{3}$ C) $\frac{k}{\sqrt{2}}$ D) $\frac{k}{\sqrt{3}}$

30. The shortest distance between the lines $\vec{r} = 3\vec{i} - 15\vec{j} + 9\vec{k} + \lambda(2\vec{i} - 7\vec{j} + 5\vec{k})$ and

$\vec{r} = (-\vec{i} + \vec{j} + 9\vec{k}) + \mu(2\vec{i} + \vec{j} - 3\vec{k})$ is

- A) $\sqrt{34}$ B) $\sqrt{3}$ C) $4\sqrt{3}$ D) $2\sqrt{3}$

31. If a plane is parallel to two vectors $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i}$ and another plane is parallel to two other vectors $\hat{i} + \hat{j}$ and $\hat{i} - \hat{k}$, then the acute angle between $2\hat{i} - \hat{j}$ and the line of intersection of the planes will be

A) $\cos^{-1}\left(\frac{3}{\sqrt{30}}\right)$ B) $\cos^{-1}\left(\frac{2}{\sqrt{30}}\right)$

C) $\cos^{-1}\left(\frac{1}{\sqrt{10}}\right)$ D) $\cos^{-1}\left(\frac{19}{30}\right)$

32. If the vector $\vec{x}$ satisfying $\vec{x} \times \vec{a} + (\vec{x} \cdot \vec{b})\vec{c} = \vec{d}$

be given by $\vec{x} = \lambda\vec{a} + \vec{a} \times \frac{\vec{d} \times (\vec{d} \times \vec{c})}{(\vec{a} \cdot \vec{c})a^2}$, then

$\lambda =$

- A) $\frac{\vec{a} \cdot \vec{c}}{a^2}$ B) $\frac{\vec{a} \cdot \vec{b}}{b^2}$ C) $\frac{\vec{c} \cdot \vec{d}}{c^2}$ D) $\frac{\vec{a} \cdot \vec{x}}{a^2}$

33. The $\perp^{lar}$ distance from the point $A(\vec{a})$ to the plane $\vec{r} \cdot \vec{n} = q$ is

A) $\left| \frac{q + \vec{a} \cdot \vec{n}}{|\vec{n}|} \right|$ B) $\left| \frac{\vec{a} \cdot \vec{n}}{|\vec{n}|} \right| + q$

C) $\left| \frac{q - \vec{a} \cdot \vec{n}}{|\vec{n}|} \right|$ D) None

34. The reflection of the point $\vec{a}$ in the plane $\vec{r} \cdot \vec{n} = q$

A) $\vec{a} + 2\left(\frac{\vec{q} - \vec{a} \cdot \vec{n}}{|\vec{n}|^2}\right)\vec{n}$ B) $\vec{a} + \left(\frac{\vec{q} - \vec{a} \cdot \vec{n}}{|\vec{n}|^2}\right)\vec{n}$

C) $\vec{a} + \left(\frac{\vec{a} \cdot \vec{n}}{|\vec{n}|^2}\right)\vec{n}$ D) None

35. The reflection of the line $\vec{r} = \vec{a} + t\vec{b}$ in the plane $\vec{r} \cdot \vec{n} = q$

A) $\vec{r} = \vec{a} + \left(\frac{\vec{q} - \vec{a} \cdot \vec{n}}{|\vec{n}|^2}\right)\vec{n} + \vec{b}$

B) $\vec{r} = \vec{a} + 2\left(\frac{q - \vec{a} \cdot \vec{n}}{|\vec{n}|^2}\right)\vec{n} + t\left(\vec{b} - 2\frac{\vec{b} \cdot \vec{n}}{|\vec{n}|^2}\vec{n}\right)$

C) $\vec{r} = \vec{a} + \vec{b} - \frac{2(\vec{b} \cdot \vec{n})}{|\vec{n}|}\vec{n}$ D) None

36. The distance of the point $P(\vec{a})$ from the plane $\vec{r} \cdot \vec{n} = q$ measured parallel to the line

$$\vec{r} = \vec{b} + t\vec{c}$$

- A) $\left| \frac{q - \vec{a} \cdot \vec{n}}{|\vec{c} \cdot \vec{n}|} \right| |\vec{c}|$ B) $\left(\frac{q^2 \vec{a} \cdot \vec{n}}{|\vec{n}|} \right) |\vec{c}|$
 C) $\frac{\vec{q} - \vec{a} \cdot \vec{n}}{|\vec{c}|}$ D) $\vec{q} + \frac{|\vec{b} - \vec{c}|}{|\vec{n}|}$

37. A square PQRS is folded along the diagonal PR so that the planes PRQ and PRS are perpendicular to one another then the shortest distance between PQ' and RS in terms of P (the length of side of the square) ($Q \rightarrow Q'$ after folding)

- A) P B) $\frac{P}{\sqrt{2}}$ C) $\sqrt{2}P$ D) $2P$

38. $\vec{OA}, \vec{OB}, \vec{OC}$ are three coterminal edges of a rectangular parallelepiped and $|\vec{OA}| = x, |\vec{OB}| = y, |\vec{OC}| = z$, then shortest distance between $\vec{OA}$ and either diagonal skew to it

- A) $\frac{yz}{\sqrt{y^2 + z^2}}$ B) $\frac{xy}{\sqrt{x^2 + y^2}}$
 C) $\frac{zx}{\sqrt{x^2 + z^2}}$ D) $\frac{z^2}{y^2}$

39. The line passing through the point $P(\vec{a})$ parallel to the line of intersection of planes

$$\vec{r} \cdot \vec{n}_1 = 1, \vec{r} \cdot \vec{n}_2 = 1$$

- A) $\vec{r} = \vec{a} + t(\vec{n}_1 + \vec{n}_2)$ B) $\vec{r} = \vec{a} + t(\vec{n}_1 \times \vec{n}_2)$
 C) $\vec{r} = \vec{a} + t(\vec{n}_1 - \vec{n}_2)$ D) $\vec{r} = t(\vec{n}_1 + \vec{n}_2)$

40. The equation of the line passing through the point $A(\vec{a})$ and is perpendicular to $\vec{r} \cdot \vec{n}_1 = q_1$ and $\vec{r} \cdot \vec{n}_2 = q_2$

- A) $\vec{r} \cdot (\vec{n}_1 \times \vec{n}_2) = q_1 + q_2$ B) $\vec{r} \cdot (\vec{n}_1 \times \vec{n}_2) = \vec{a} \cdot (\vec{n}_1 \times \vec{n}_2)$
 C) $\vec{r} \cdot (\vec{n}_1 \times \vec{n}_2) = \vec{a} \cdot (\vec{n}_1 + \vec{n}_2)$
 D) $\vec{r} \cdot (\vec{n}_1 + \vec{n}_2) = q_1 + q_2$

41. If $\vec{p}, \vec{q}, \vec{r}$ are three non coplanar vectors then

$$\begin{vmatrix} \vec{p} & \vec{q} & \vec{r} \\ \vec{p} \cdot \vec{a} & \vec{q} \cdot \vec{a} & \vec{r} \cdot \vec{a} \\ \vec{p} \cdot \vec{b} & \vec{q} \cdot \vec{b} & \vec{r} \cdot \vec{b} \end{vmatrix} \frac{1}{\sqrt{[\vec{p} \times \vec{q} \quad \vec{q} \times \vec{r} \quad \vec{r} \times \vec{p}]}} =$$

- A) $\vec{a} \times \vec{b}$ B) $\vec{a} + \vec{b}$ C) $\vec{a} - \vec{b}$ D) $\vec{0}$

42. The three planes $\vec{r} \cdot \vec{n}_1 = p_1, \vec{r} \cdot \vec{n}_2 = p_2,$

$\vec{r} \cdot \vec{n}_3 = p_3$ have a common line of intersection then

$$p_1(\vec{n}_2 \times \vec{n}_3) + p_2(\vec{n}_3 \times \vec{n}_1) + p_3(\vec{n}_1 \times \vec{n}_2) =$$

- A) $\vec{n}_1 + \vec{n}_2 + \vec{n}_3$ B) $\vec{0}$
 C) $\vec{n}_1 - \vec{n}_2 + \vec{n}_3$ D) $\vec{n}_1 + 2\vec{n}_2 + 3\vec{n}_3$

43. Let v be the volume of parallelepiped formed by the vectors

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

$$\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$$

$$\vec{c} = c_1\vec{i} + c_2\vec{j} + c_3\vec{k}$$

If a_r, b_r, c_r for $r = 1, 2, 3$ are non negative

reals and $\sum_{r=1}^3 (a_r + b_r + c_r) = 3L$ then

- A) $v \leq L^3$ B) $v \geq L^3$
 C) $v \geq 2L + 3$ D) $v = L$

44. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ and

$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$ then maximum value of θ is

- A) $\frac{\pi}{2}$ B) $\frac{\pi}{4}$ C) $\frac{2\pi}{3}$ D) $\frac{\pi}{6}$

45. If $\vec{a}, \vec{b}, \vec{c}$ are three non-zero, non-coplanar

vectors and $\vec{b}_1 = \vec{b} - \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}, \vec{b}_2 = \vec{b} + \frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \vec{a},$

$$\vec{c}_1 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}, \quad \vec{c}_2 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}_1}{|\vec{b}_1|^2} \vec{b}_1,$$

$$\vec{c}_3 = \vec{c} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}_2}{|\vec{b}_2|^2} \vec{b}_2, \quad \vec{c}_4 = \vec{a} - \frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a}$$

Then which of the following is a set of mutually orthogonal vectors [IIT JEE 2005]

(A) $\{\vec{a}, \vec{b}_1, \vec{c}_1\}$ (B) $\{\vec{a}, \vec{b}_1, \vec{c}_2\}$

(C) $\{\vec{a}, \vec{b}_2, \vec{c}_3\}$ (D) $\{\vec{a}, \vec{b}_2, \vec{c}_4\}$

46. The edges of a parallelopiped are of unit length and are parallel to non-coplanar unit

vector $\hat{a}, \hat{b}, \hat{c}$ such that $\hat{a} \cdot \hat{b} = \hat{b} \cdot \hat{c} = \hat{c} \cdot \hat{a} = \frac{1}{2}$.

Then, the volume of the parallelopiped is [IIT JEE 2008]

(A) $\frac{1}{\sqrt{2}}$ cu unit (B) $\frac{1}{2\sqrt{2}}$ cu unit

(C) $\frac{\sqrt{3}}{2}$ cu unit (D) $\frac{1}{\sqrt{3}}$ cu unit

47. Vectors $\vec{a}, \vec{b}$ are non-zero, non-collinear vectors such that $|\vec{a}| = 2$, $\vec{a} \cdot \vec{b} = 1$ and angle

between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$, if $\vec{r}$ is any vector

satisfying $\vec{r} \cdot \vec{a} = 2$, $\vec{r} \cdot \vec{b} = 8$,

$(\vec{r} + 2\vec{a} - 10\vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3}$ also

$(\vec{r} + 2\vec{a} - 10\vec{b}) = \lambda(\vec{a} \times \vec{b})$ then $\lambda =$

A) $\frac{4}{\sqrt{3}}$ B) $\frac{2}{\sqrt{3}}$ C) $\sqrt{3}$ D) $\frac{1}{\sqrt{3}}$

48. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$ be three vectors. A vector $\vec{v}$ in the plane of $\vec{a}$ and $\vec{b}$, whose projection on

$\vec{c}$ is $\frac{1}{\sqrt{3}}$, is given by [IIT 2011]

A) $\hat{i} - 3\hat{j} + 3\hat{k}$ B) $-3\hat{i} - 3\hat{j} + \hat{k}$

C) $3\hat{i} - \hat{j} + 3\hat{k}$ D) $\hat{i} + 3\hat{j} - 3\hat{k}$

MULTIANSWER QUESTIONS

49. A particle in equilibrium is subjected to four

forces $\vec{F}_1 = -10\hat{k}, \vec{F}_2 = u \left[\frac{4}{13}\hat{i} - \frac{12}{13}\hat{j} + \frac{3}{13}\hat{k} \right],$

$\vec{F}_3 = V \left[-\frac{4}{13}\hat{i} - \frac{12}{13}\hat{j} + \frac{3}{13}\hat{k} \right],$

$\vec{F}_4 = W \left[\cos \theta \hat{i} + \sin \theta \hat{j} \right],$ then

(A) $U = \frac{65}{3}(1 - 3 \cot \theta)$

(B) $V = \frac{65}{3}(1 + 3 \cot \theta)$

(C) $W = 40 \operatorname{cosec} \theta$

(D) $U + V = \frac{130}{3}$

50. In quadrilateral ABCD, $\vec{AB} = \vec{a}, \vec{BC} = \vec{b}, \vec{AD} = \vec{b} - \vec{a}$. If M is the midpoint of BC and N is point on DM such

that $\vec{DN} = \frac{4}{5} \vec{DM}$, then

(A) $5\vec{AN} = 3\vec{AC}$ (B) $3\vec{AN} = 5\vec{AC}$

(C) $5\vec{AN} - 3\vec{AC} = \vec{0}$ (D) $3\vec{AN} + 5\vec{AC} = \vec{0}$

51. Let $\vec{p} = 4\vec{q} + 3\vec{r}$ and $3\vec{p} = 2\vec{q} - 3\vec{r}$, then

(A) $\vec{p}, \vec{q}$ are having same directions, $|\vec{p}| > |\vec{q}|$

(B) $\vec{p}, \vec{q}$ are having opposite directions, $|\vec{p}| > |\vec{q}|$

(C) $\vec{p}, \vec{r}$ are having same directions $|\vec{p}| > |\vec{r}|$

(D) $\vec{p}, \vec{r}$ are having opposite directions $|\vec{p}| > |\vec{r}|$

52. Let $\vec{a} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}; \vec{b} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}; \vec{c} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. then the numbers α, β, γ such that

$$\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = \begin{bmatrix} -2 \\ -5 \\ 6 \end{bmatrix}, \text{ are}$$

- (A) $\alpha = -1$ (B) $\beta = -2$
 (C) $\gamma = 3$ (D) $\alpha + \beta + \gamma = 0$

53. The vector $\vec{a} + \vec{b}$ bisects the angle between the vectors $\vec{a}$ and $\vec{b}$ if

- (A) $|\vec{a}| = |\vec{b}|$
 (B) angle between $\vec{a}$ and $\vec{b}$ is zero
 (C) $|\vec{a}| + |\vec{b}| = 0$
 (D) none of the above

54. If each of $\vec{a}, \vec{b}, \vec{c}$ is orthogonal to sum of other two vectors and $|\vec{a}| = 3, |\vec{b}| = 4, |\vec{c}| = 5$

then

- A) when $\vec{a}$ is equally inclined with coordinate axes then $\tan \theta = \pm\sqrt{2}$
 B) range of $|\vec{a} - \vec{b}|$ is $[1, 7]$
 C) range of $|\vec{b} - \vec{c}|$ is $[1, 9]$
 D) None of the above

55. The vectors $\vec{a}, \vec{b}, \vec{c}$ are of the same length and the angle between any two of them is the same. If $\vec{a} = \hat{i} + \hat{j}, \vec{b} = \hat{j} + \hat{k}$, then $\vec{c}$ is

- A) $-\frac{1}{3}(\hat{i} - 4\hat{j} + \hat{k})$ B) $\hat{i} + \hat{k}$
 C) $\frac{1}{3}(-\hat{i} + 4\hat{j} + \hat{k})$ D) $(\hat{i} + 2\hat{j} + 3\hat{k})$

56. ABCD is a square described in the anticlockwise direction. The position vector of A is $\vec{i} - 2\vec{j}$ and the position vector of centre of square is $6\vec{i} - \vec{j}$, then

- A) position vector of B is $7\vec{i} - 6\vec{j}$
 B) position vector of D is $5\vec{i} + 4\vec{j}$
 C) position vector of C is $10\vec{i} + \vec{j}$
 D) area of square is 52 sq.units

57. The values of x for which the angle between the vectors $\vec{a} = x\hat{i} - 3\hat{j} - \hat{k}$ and $\vec{b} = 2x\hat{i} + x\hat{j} - \hat{k}$ is acute, and the angle between the vector $\vec{b}$ and the axis of ordinates is obtuse are

- (A) 1, 2 (B) -2, -3
 (C) all $x < 0$ (D) all $x > 0$

58. Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b}$ be two vectors perpendicular to each other in xy-plane. The vectors $\vec{c}$ in the same plane having projections 1 and 2 along $\vec{a}$ and $\vec{c}$ are

- (A) $-\frac{2}{3}\hat{i} + \frac{11}{2}\hat{j}$ (B) $2\hat{i} - \hat{j}$
 (C) $-\frac{2}{5}\hat{i} + \frac{11}{5}\hat{j}$ (D) $\frac{2}{3}\hat{i} + \frac{11}{2}\hat{j}$

59. The vectors $\vec{AB} = 3\hat{i} - 2\hat{j} + 2\hat{k}$ and $\vec{BC} = -\hat{i} - 2\hat{k}$ are the adjacent sides of a parallelogram. The angle between its diagonals is

- (A) $\pi/4$ (B) $\pi/3$ (C) $3\pi/4$ (D) $2\pi/3$

60. The position vectors of the vertices A, B, C of a tetrahedron ABCD are $\vec{i} + \vec{j} + \vec{k}, \vec{i}$ and $3\vec{i}$ respectively. The altitude from the vertex D to the opposite face ABC meet the median line through A of the $\triangle ABC$ at E. If the length of side AD is 4 and volume of

tetrahedron is $\frac{2\sqrt{2}}{3}$ then P.V of E is

- A) $-\vec{i} + 3\vec{j} + 3\vec{k}$ B) $3\vec{i} - \vec{j} - \vec{k}$
 C) $\vec{i} - 3\vec{j} + \vec{k}$ D) $\vec{i} - \vec{j} - \vec{k}$

61. The vectors $(x, x+1, x+2), (x+3, x+4, x+5)$ and $(x+6, x+7, x+8)$ are coplanar for

- (A) all values of x (B) $x < 0$
 (C) $x > 0$ (D) none of these

62. A pyramid with vertex at the point P, whose position vector is $4\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}$ has a regular hexagonal base ABCDEF. Position vectors of points A and B are $\vec{i}$ and $\vec{i} + 2\vec{j}$ respectively. Centre of the base has the position vector $\vec{i} + \vec{j} + \sqrt{3}\vec{k}$. Altitude drawn from P on the base meets the diagonal AD at point G and volume of the pyramid is

$6\sqrt{3}$ cubic units then all possible position vectors of G are

- A) $\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}$
 B) $-(\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k})$
 C) $3\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}$
 D) $-3\vec{i} + 2\vec{j} + 2\sqrt{3}\vec{k}$

63. If $\vec{a}, \vec{b}$ are non collinear vectors and $\vec{r} + \vec{r} \times \vec{a} = \vec{b}$ then

- A) $|\vec{r}| = \frac{\sqrt{(\vec{a} \cdot \vec{b})^2 + \vec{b}^2}}{1 + \vec{a}^2}$
 B) $\vec{r} = \frac{(\vec{a} \cdot \vec{b})\vec{a} + \vec{b} + \vec{a} \times \vec{b}}{1 + |\vec{a}|^2}$
 C) $\vec{r} = \frac{\vec{b} + \vec{a} \times \vec{b}}{1 + |\vec{a}|^2}$
 D) $|\vec{r}| = \frac{|\vec{b} + \vec{a} \times \vec{b}|}{1 + |\vec{a}|^2}$

64. Unit vectors $\vec{a}$ and $\vec{b}$ are perpendicular and unit vector $\vec{c}$ is inclined at an angle θ to both $\vec{a}$ and $\vec{b}$. If $\vec{c} = \alpha\vec{a} + \beta\vec{b} + \gamma(\vec{a} \times \vec{b})$ then

- A) $\alpha = \beta$ B) $\gamma^2 = 1 - 2\alpha^2$
 C) $\gamma^2 = -\cos 2\theta$ D) $\beta^2 = \frac{1 + \cos 2\theta}{2}$

65. $\vec{a}$ and $\vec{b}$ are two given vectors. With these vectors as adjacent sides, a parallelogram is constructed. The vector which is the altitude of the parallelogram and which is perpendicular to $\vec{a}$ is

- A) $\frac{(\vec{a} \cdot \vec{b})}{|\vec{a}|^2} \vec{a} - \vec{b}$
 B) $\frac{1}{|\vec{a}|^2} \{ |\vec{a}|^2 \vec{b} - (\vec{a} \cdot \vec{b}) \vec{a} \}$

C) $\frac{\vec{a} \times (\vec{a} \times \vec{b})}{|\vec{a}|^2}$ D) $\frac{\vec{a} \times (\vec{b} \times \vec{a})}{|\vec{b}|^2}$

66. Let $\vec{r}$ be a unit vector satisfying $\vec{r} \times \vec{a} = \vec{b}$, where $|\vec{a}| = \sqrt{3}$ and $|\vec{b}| = \sqrt{2}$ then

- A) $\vec{r} = \frac{2}{3}(\vec{a} + \vec{a} \times \vec{b})$ B) $\vec{r} = \frac{1}{3}(\vec{a} + \vec{a} \times \vec{b})$
 C) $\vec{r} = \frac{2}{3}(\vec{a} - \vec{a} \times \vec{b})$ D) $\vec{r} = \frac{1}{3}(-\vec{a} + \vec{a} \times \vec{b})$

67. If $\vec{a}$ and $\vec{b}$ are two unit vectors perpendicular to each other and $\vec{c} = \lambda_1\vec{a} + \lambda_2\vec{b} + \lambda_3(\vec{a} \times \vec{b})$, then which of the following is(are) true?

- A) $\lambda_1 = \vec{a} \cdot \vec{c}$ B) $\lambda_2 = |\vec{b} \times \vec{c}|$
 C) $\lambda_3 = |(\vec{a} \times \vec{b}) \times \vec{c}|$
 D) $\lambda_1 + \lambda_2 + \lambda_3 = (\vec{a} + \vec{b} + \vec{a} \times \vec{b}) \cdot \vec{c}$

68. If the side $\overline{AB}$ of an equilateral triangle, ABC lying in the xy-plane is $3\hat{i}$, then the side $\overline{CB}$ can be

- A) $-\frac{3}{2}(\hat{i} - \sqrt{3}\hat{j})$ B) $\frac{3}{2}(\hat{i} - \sqrt{3}\hat{j})$
 C) $-\frac{3}{2}(\hat{i} + \sqrt{3}\hat{j})$ D) $\frac{3}{2}(\hat{i} + \sqrt{3}\hat{j})$

69. Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-coplanar vectors and $\vec{d}$ be a non-zero vector perpendicular to $(\vec{a} + \vec{b} + \vec{c})$. Now $\vec{d} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$ then

- A) $\frac{\vec{d} \cdot (\vec{a} + \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]} = 2$ B) $\frac{\vec{d} \cdot (\vec{a} + \vec{c})}{[\vec{a} \ \vec{b} \ \vec{c}]} = -2$

C) minimum value of $x^2 + y^2$ is $\frac{\pi^2}{4}$

D) minimum value of $x^2 + y^2$ is $\frac{5\pi^2}{4}$

70. The scalars l and m such that $l\vec{a} + m\vec{b} = \vec{c}$, where $\vec{a}, \vec{b}$ and $\vec{c}$ are given vectors, are equal to

A) $l = \frac{(\vec{c} \times \vec{b}) \cdot (\vec{a} \times \vec{b})}{(\vec{a} \times \vec{b})^2}$

B) $l = \frac{(\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{a})}{|\vec{b} \times \vec{a}|}$

C) $m = \frac{(\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{a})}{(\vec{b} \times \vec{a})^2}$

D) $m = \frac{(\vec{c} \times \vec{a}) \cdot (\vec{b} \times \vec{a})}{|\vec{b} \times \vec{a}|}$

71. If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are any four vectors then

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is a vector :

- (A) perpendicular to $\vec{a}, \vec{b}, \vec{c}, \vec{d}$
 (B) along the line of intersection of two planes, one containing $\vec{a}, \vec{b}$ and the other containing $\vec{c}, \vec{d}$
 (C) equally inclined to both $\vec{a} \times \vec{b}$ and $\vec{c} \times \vec{d}$
 (D) coplanar with $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$

ASSERTION-REASONING QUESTIONS

In the following set of questions, a Statement I is given and a corresponding Statement II is given just below it. Mark the correct answer as:

- (A) Both Statement I and II are true and Statement II is a correct explanation for Statement I.
 (B) Both Statement I and II are true but Statement II is not a correct explanation for Statement I.

(C) Statement I is true and Statement II is false

(D) Statement I is false and Statement II is true

72. Statement 1: In $\triangle ABC$, $\vec{AB} + \vec{BC} + \vec{CA} = 0$

Statement 2: If $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, then $\vec{AB} = \vec{a} + \vec{b}$

73. Statement 1: If $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then $\vec{a}$ and $\vec{b}$ are perpendicular to each other

Statement 2: If the diagonals of a parallelogram are equal in magnitude, then the parallelogram is a rectangle

74. Statement 1: Let $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ be three points such that $\vec{a} = 2\hat{i} + \hat{k}, \vec{b} = 3\hat{i} - \hat{j} + 3\hat{k}, \vec{c} = -\hat{i} + 7\hat{j} - 5\hat{k}$. Then $OABC$ is a tetrahedron

Statement 2: Let $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ be three points such that $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar. Then $OABC$ is a tetrahedron, where O is the origin

75. Statement 1: Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ be the position vectors of four points A, B, C and D and $3\vec{a} - 2\vec{b} + 5\vec{c} - 6\vec{d} = \vec{0}$. Then points A, B, C and D are coplanar

Statement 2: Three non-zero, linearly dependent coinitial vectors $(\vec{PQ}, \vec{PR}$ and $\vec{PS})$ are coplanar

76. Statement 1: If $\vec{a}, \vec{b}$ are sides of parallelogram and $|\vec{a}| = 3, |\vec{b}| = 4$ and $|\vec{a} + \vec{b}| = 5$, then $|\vec{a} - \vec{b}| = 5$.

Statement 2: The length of the diagonals of a rectangle is the same

77. Statement I: If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, then $\vec{b} \times \vec{c}, \vec{c} \times \vec{a}$ and $\vec{a} \times \vec{b}$ are also non-coplanar

Statement II: Vectors $\vec{a}, \vec{b}, \vec{c}$ are coplanar

if and only if $[\vec{a}, \vec{b}, \vec{c}] = 0$

78. **Statement I:** If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors,

then
$$\begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = 0$$

Statement II:

$$[\vec{a} \vec{b} \vec{c}](\vec{a} \times \vec{b}) = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

79. **Statement I:** The shortest distance between

the lines $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ and

$$\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1} \text{ is } 2\sqrt{19}$$

Statement II: The shortest distance between two skew lines $\vec{r} = \vec{a} + t\vec{b}$ and $\vec{r} = \vec{c} + s\vec{d}$

is
$$\frac{|(\vec{a} - \vec{c}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|}$$

80. **Statement I:** If any ΔABC , $\cos 2A + \cos 2B$

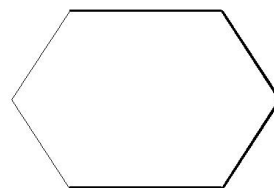
$$+ \cos 2C \geq -\frac{3}{2}$$

Statement II: If O is the circum centre of ΔABC , then $|\vec{OA} + \vec{OB} + \vec{OC}| \geq 0$.

81. $\vec{PQ}, \vec{QR}, \vec{RS}, \vec{ST}, \vec{TU}$ and $\vec{UP}$ represent the sides of a regular hexagon

Statement I: $\vec{PQ} \times (\vec{RS} + \vec{ST}) \neq 0$

Statement II: $\vec{PQ} \times \vec{RS} = 0$ and $\vec{PQ} \times \vec{ST} \neq 0$



82. Consider the planes $3x - 6y - 2z = 15$ and $2x + y - 2z = 5$.

Statement I: Let $\vec{n}_1 = (3, -6, -2)$ and $\vec{n}_2 = (2, 1, -2)$ be normals to the planes then $\vec{n}_1 \times \vec{n}_2$ is parallel to the line of intersection of the planes.

Statement II: The vector $14\vec{i} + 2\vec{j} + 15\vec{k}$ is parallel to the line of intersection of the planes.

83. **Statement -1 :** If the vector $\vec{a}$ and $\vec{c}$ are non-collinear, then the lines

$$\vec{r} = 6\vec{a} - \vec{c} + \lambda(2\vec{c} - \vec{a}) \text{ and}$$

$$\vec{r} = \vec{a} - \vec{c} + \mu(\vec{a} + 3\vec{c}) \text{ are coplanar}$$

Statement -2 : There exist λ and μ such that the two values of $\vec{r}$ becomes same.

84. **Statement -1 :** If $\vec{u}$ and $\vec{v}$ are unit vectors inclined at an angle α and $\vec{x}$ is a unit vector bisecting the angle between them, then

$$\vec{x} = \frac{\vec{u} + \vec{v}}{2 \cos \frac{\alpha}{2}}$$

Statement -2 : If ABC is an isosceles triangle with $AB=AC = 1$, then vector representing bisector of Angle A is given by

$$\vec{AD} = \frac{\vec{AB} + \vec{AC}}{2}.$$

85. **Statement 1 :**

$$[2\vec{a} + 3\vec{b}, 3\vec{b} + 5\vec{c}, 5\vec{c} + 7\vec{a}] = 25[\vec{a} \vec{b} \vec{c}]$$

because

Statement 2 : If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar

$[\vec{a} \vec{b} \vec{c}]$ is non zero

86. **Statement 1 :** If $\vec{a}, \vec{b}, \vec{c}$ are non coplanar then $\vec{a}, \vec{a} + \vec{b}, \vec{a} + \vec{b} + \vec{c}$ are also non coplanar because

Statement 2 :

$$[\vec{a}, \vec{a} + \vec{b}, \vec{a} + \vec{b} + \vec{c}] = [\vec{a} \vec{b} \vec{c}]$$

PASSAGE QUESTIONS

- (P) A tetrahedron whose edges are all equal in length is regular tetrahedron. The faces of regular tetrahedron are equilateral triangle and angle between any two concurrent edge of a regular tetrahedron is 60° then answer the following questions by taking length of the edge as a

87. Angle between any edge and a face not containing the edge is

A) $\cos^{-1}\left(\frac{1}{3}\right)$ B) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
C) $\cos^{-1}\left(\frac{2}{3}\right)$ D) 90°

88. Angle between any two opposite edges is

A) 60° B) 90°
C) $\cos^{-1}\left(\frac{1}{3}\right)$ D) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

89. The distance of any vertex from opposite face is

A) $\sqrt{\frac{2}{3}}a$ B) $\frac{a}{\sqrt{3}}$ C) $\frac{a}{2}$ D) $\frac{a}{2\sqrt{3}}$

- (P) If $\vec{a}, \vec{b}, \vec{c}$ be non coplanar unit vectors equally inclined to one another at an angle θ and if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} = p\vec{a} + q\vec{b} + r\vec{c}$ then

90. $p = \dots\dots\dots$

(A) $\frac{1}{\sqrt{2\cos\theta}}$ (B) $\frac{1}{\sqrt{1+2\cos\theta}}$

(C) $\frac{1}{\sqrt{1-\cos\theta}}$ (D) $\frac{1}{\sqrt{1-2\cos\theta}}$

91. $r = \dots\dots\dots$

(A) $\frac{1}{\sqrt{2\cos\theta}}$ (B) $\frac{1}{\sqrt{2\sin\theta}}$

(C) $\frac{1}{\sqrt{1+2\cos\theta}}$ (D) $\frac{1}{\sqrt{1-2\cos\theta}}$

92. $q = \dots\dots\dots$

(A) $\frac{-2\cos\theta}{\sqrt{1+2\cos\theta}}$ (B) $\frac{-2\sin\theta}{\sqrt{1+2\cos\theta}}$

(C) $\frac{-2\cos\theta}{\sqrt{1+\cos\theta}}$ (D) $\frac{1}{\sqrt{1+2\cos\theta}}$

- (P) Let $\vec{u}, \vec{v}, \vec{w}$ be a 3 unit vectors such that

$$\vec{u} + \vec{v} + \vec{w} = \vec{a}, \quad \vec{u} \times (\vec{v} \times \vec{w}) = \vec{b}$$

$$(\vec{u} \times \vec{v}) \times \vec{w} = \vec{c}, \quad \vec{a} \cdot \vec{u} = \frac{3}{2}, \quad \vec{a} \cdot \vec{v} = \frac{7}{4},$$

$$|\vec{a}| = 2 \text{ then}$$

93. $\vec{u} = \dots\dots\dots$

(A) $\vec{a} + \frac{4}{3}\vec{b} + \frac{8}{3}\vec{c}$ (B) $\vec{a} + \vec{b} + \vec{c}$

(C) $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$ (D) $\vec{a} + \frac{\vec{b}}{2} + \frac{\vec{c}}{3}$

94. $\vec{v} = \dots\dots\dots$

(A) $\vec{a} + \vec{b} + \vec{c}$ (B) $\frac{\vec{a} + \vec{b}}{2} + \vec{c}$

(C) $-4\vec{c}$ (D) $\vec{a} + \frac{\vec{b}}{2} + \frac{\vec{c}}{3}$

95. $\vec{w} = \dots\dots\dots$

(A) $\frac{\vec{a} + \vec{b}}{3}$ (B) $\frac{\vec{c} - \vec{b}}{3}$

(C) $\left(\frac{\vec{a} + \vec{b}}{3}\right)4$ (D) $\frac{4}{3}(\vec{c} - \vec{b})$

VECTOR ALGEBRA

JEE ADVANCED - VOL - I

(P) If $\vec{u} + \vec{v} + \vec{w} = \vec{a}$, $\vec{u} \times \vec{v} = \vec{b}$, $\vec{v} \times \vec{w} = \vec{c}$,
 $\vec{a} \cdot \vec{u} = \vec{a} \cdot \vec{v} = 1$ and $|\vec{a}| = \sqrt{3}$ then

96. $\vec{u} =$ _____
 A) $\frac{\vec{a} + \vec{a} \times \vec{c} + \vec{a} \times \vec{b}}{3}$ B) $\frac{\vec{a} \times \vec{c}}{3}$
 C) $\frac{\vec{a}}{3}$ D) $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$

97. $\vec{v} =$ _____
 A) $\frac{\vec{a} - \vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$
 B) $\frac{\vec{a} + \vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$
 C) $\frac{\vec{a} \times \vec{b}}{3}$ B) None

98. $\vec{w} =$ _____
 A) $\frac{\vec{a} - \vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$ B) $\frac{\vec{a} - 2\vec{a} \times \vec{c} - \vec{a} \times \vec{b}}{3}$
 C) $\frac{\vec{a} + \vec{a} \times \vec{b} + \vec{a} \times \vec{c}}{3}$ D) None

MATRIX MATCHING QUESTIONS

99. COLUMN-I

(A) The projection of the vector $\vec{i} + \vec{j} + \vec{k}$ On the line whose vector equation is $\vec{r} = (3+t)\vec{i} + (2t-1)\vec{j} + 3t\vec{k}$, t being the scalar parameter, is

(B) Let $\vec{AB} = 3\vec{i} + \vec{j} - \vec{k}$ and $\vec{AC} = \vec{i} - \vec{j} + 3\vec{k}$. If the point P on the line segment BC is equidistant

from AB and AC then $|\vec{AP}|$ is

(C) The distance between the point with position vector $-\vec{i} - 5\vec{j} - 10\vec{k}$ and the point of intersection of the line

$\vec{r} = 2\vec{i} - \vec{j} + 2\vec{k} + t(3\vec{i} + 4\vec{j} + 12\vec{k})$ with the plane $\vec{r} \cdot (\vec{i} - \vec{j} + \vec{k}) = 5$ is

(D) The perpendicular distance from the origin to the plane through the point $7\vec{j} - 7\vec{k}$ and containing the line

$\vec{r} = (-\vec{i} + 3\vec{j} - 2\vec{k}) + t(-3\vec{i} + 2\vec{j} + \vec{k})$ is

COLUMN-II

(P) $\sqrt{5}$

(Q) $\frac{6}{\sqrt{14}}$

(R) 0

(S) 13

100. COLUMN-I

(A) The distance of the point (1,0,1) from the line $\vec{r} = (\vec{i} + 2\vec{j} + 4\vec{k}) + \lambda(\vec{i} - \vec{k})$ is

(B) If A denote the image of origin in the line $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{1}$ then OA =

(C) If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-coplanar unit vectors, each inclined with other at an angle 30° , then volume of tetrahedron whose edges are $\vec{a}, \vec{b}$ and $\vec{c}$ is

(D) If $\vec{a}, \vec{b}$ and $\vec{c}$ are unit vectors such that

$\vec{a} \cdot \vec{b} = 0, (\vec{a} - \vec{c}) \cdot (\vec{b} + \vec{c}) = 0$ and

$\vec{c} = \lambda \vec{a} + \mu \vec{b} + \omega(\vec{a} \times \vec{b})$ where λ, μ and ω

are scalars then $2\mu(\mu+1) + \omega^2 =$

COLUMN-II

(p) $\frac{1}{2}\sqrt{3\sqrt{3}-5}$

(q) 0

(r) $\sqrt{\frac{17}{2}}$

(s) $\sqrt{\frac{106}{9}}$

101. COLUMN-I(A) The position vectors of vertices of ΔABC are

$\vec{a}, \vec{b}, \vec{c}$ and $\vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c} = 3$,

 $[\vec{a}, \vec{b}, \vec{c}] = 0$, the position vector of the orthocentre of ΔABC is(B) If $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors and $\vec{a}$ is not parallel to $\vec{b}$ then

$$\left\{ \left(\vec{c} \times \vec{b} \right) \cdot \left(\vec{a} \times \vec{b} \right) \right\} \vec{a} + \left\{ \left(\vec{a} \times \vec{c} \right) \cdot \left(\vec{a} \times \vec{b} \right) \right\} \vec{b}$$
 is equal to

(C) Let $\vec{x}, \vec{y}$ and $\vec{z}$ be unit vectors such that

$$\vec{x} + \vec{y} + \vec{z} = \vec{a}, \quad \vec{x} \times (\vec{y} + \vec{z}) = \vec{b},$$

$$(\vec{x} \times \vec{y}) \times \vec{z} = \vec{c}, \quad \vec{a} \cdot \vec{x} = 3, \vec{a} \cdot \vec{y} = \frac{7}{4}$$

and $|\vec{a}| = 2$ then $\vec{x} =$

(D) Let $\vec{a}, \vec{b}$ and $\vec{c}$ be three non-zero vectors, no two of which are collinear. If the vector

$$3\vec{a} + 7\vec{b}$$
 is collinear with $\vec{c}$ and $3\vec{b} + 2\vec{c}$ is

collinear with $\vec{a}$, then $21\vec{b} + 14\vec{c} =$

COLUMN-II

(p) $\frac{1}{3}(3\vec{a} + 4\vec{b} + 8\vec{c})$

(q) $-9\vec{a}$

(r) $\left\{ \left(\vec{a} \times \vec{b} \right) \cdot \left(\vec{a} \times \vec{b} \right) \right\} \vec{c}$

(s) $\vec{a} + \vec{b} + \vec{c}$

INTEGER QUESTIONS

102. E and F are the points on the sides BC and CD of a parallelogram ABCD. If $\vec{BE} = 4 \times \vec{EC}$ and $\vec{CF} = 4 \times \vec{FD}$ the line EF cuts diagonal AC in G. If $\vec{AG} = \lambda \vec{AC}$ then the value $\frac{25\lambda}{7}$

103. Given 2006 vectors on a plane, sum of every 2005 vectors is a scalar multiple of the other vector, not all vectors are scalar multiple of each other then magnitude of sum of all the vectors is

104. The distance of the point $B(\hat{i} + 2\hat{j} + 3\hat{k})$ to the line passing through $A(4\hat{i} + 2\hat{j} + 3\hat{k})$ and which is parallel to the vector $C(2\hat{i} + 3\hat{j} + 6\hat{k})$ is λ , then $\frac{49\lambda^2}{81} =$

105. Let $\vec{V}_1 = \hat{i} + \hat{j} + \hat{k}$; $\vec{V}_2 = a\hat{i} + b\hat{j} + c\hat{k}$ where $a, b, c \in \{-2, -1, 0, 1, 2\}$, if n be the number of non zero vectors such that $\vec{V}_2$ is perpendicular to $\vec{V}_1$ then $\frac{n}{3}$ is

106. $\vec{a}, \vec{b}, \vec{c}$ are 3 unit vectors such that each is inclined at an angle θ with the other. A unit vector $\vec{d}$ is equally inclined with these vectors at an angle α , then $4\cos\theta - 3\cos 2\alpha$ is

107. Let $\vec{c}$ be a unit vector coplanar with $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$ such that $\vec{c}$ is perpendicular to $\vec{a}$. If P be the projection of $\vec{c}$ along $\vec{b}$ then the value of $\frac{\sqrt{11}}{p}$ is.

VECTOR ALGEBRA

JEE ADVANCED - VOL - I

108. If θ is the acute angle between the medians drawn through the acute angle of an isosceles right angled triangle then the value of $4 \sec \theta$ is
109. The resultant of vectors $\vec{a}$ and $\vec{b}$ is $\vec{c}$. If $\vec{c}$ trisects the angle between $\vec{a}$ and $\vec{b}$, $|\vec{a}| = 6, |\vec{b}| = 4$, then $|\vec{c}|$ is
110. Let $\vec{V}_1 = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{V}_2 = \hat{i} - 2\hat{j} + \hat{k}$, $\vec{V}_3 = -2\hat{i} + 2\hat{j} + \hat{k}$ are three vectors. Let $\vec{V}$ be a vector such that it can be expressed as a linear combination of $\vec{V}_1$ and $\vec{V}_2$ also $\vec{V} \cdot \vec{V}_3 = 0$ and the projection of vector $\vec{V}$ on $\hat{i} - \hat{j} + \hat{k}$ is $6\sqrt{3}$. If $\vec{V} = t(\hat{i} + 3\hat{j} - 4\hat{k})$ then the absolute value of 't' is
111. If the projection of $A(\vec{a})$ in the plane $\vec{r} \cdot \hat{n} = 2$ is $\hat{n} \times (\vec{a} \times \hat{n}) + \vec{b}$ and $|\vec{a}| = 3$ then $|\vec{b}| =$
112. $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{c} = \hat{i} + x_1\hat{j} + x_2\hat{k}$ are linearly dependent, $|\vec{c}| = 2$ then (i) $x_1 =$ (ii) $x_2 =$
113. Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors having magnitudes 1, 2, 3 respectively satisfy the relation $|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}| = 6$
- If $\vec{d}$ is a unit vector coplanar with $\vec{b}$ and $\vec{c}$ such that $\vec{b} \cdot \vec{d} = 1$ then the value of $|(\vec{a} \times \vec{c}) \cdot \vec{d}|^2 + |(\vec{a} \times \vec{c}) \times \vec{d}|^2$ is
114. If $\vec{a}, \vec{b}, \vec{c}$ are three non coplanar, non zero vectors and $\vec{r}$ is any vector in space such that $(\vec{a} \times \vec{b}) \times (\vec{r} \times \vec{c}) + (\vec{b} \times \vec{c}) \times (\vec{r} \times \vec{a}) + (\vec{c} \times \vec{a}) \times (\vec{r} \times \vec{b}) = \lambda [\vec{a} \vec{b} \vec{c}] \vec{r}$ then λ is
115. The position vectors of points A, B, C, D and E are respectively $\hat{i} + 2\hat{j} + \hat{k}, 2\hat{i} + \hat{j} + 2\hat{k}, -4\hat{j} + 4\hat{k}, 2\hat{i} - 2\hat{j} + 2\hat{k}$ and $4\hat{i} + \hat{j} + 2\hat{k}$. If the line AB cuts the plane CDE at point P having position vector $\alpha \vec{i} + \beta \vec{j} + \gamma \vec{k}$ then the value of $(\alpha + \beta + \gamma)$ is
116. If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i} + \hat{j} + 3\hat{k}}{\sqrt{14}}$, then the value of $(2\vec{a} + \vec{b}) \cdot [(\vec{a} \times \vec{b}) \times (\vec{a} - 2\vec{b})]$ is [IIT JEE 2010]
117. Let $\vec{a}, \vec{b}, \vec{c}$ be coplanar unit vectors such that $\vec{b} \cdot \vec{c} = \cos \alpha, \vec{c} \cdot \vec{a} = \cos \beta, \vec{a} \cdot \vec{b} = \cos \gamma$ then the value of $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma - 2 \cos \alpha \cos \beta \cos \gamma$ is
118. If the vector $-\hat{i} + \hat{c}\hat{j} + b\hat{k}, -\hat{c}\hat{i} + \hat{j} + a\hat{k}, b\hat{i} + a\hat{j} - \hat{k}$ are coplanar vectors and $|a| \leq 1$, then the maximum value of $|a\hat{i} + b\hat{j} + c\hat{k}|^2$ is
119. Given a tetrahedron DABC with AB = 12, CD = 6. If the shortest distance between the skew lines AB and CD is 8 and the angle between them is $\pi/6$, then the $\frac{1}{6}$ of the volume of the tetrahedron is
120. The lengths of two opposite edges of tetrahedron are a and b and their S.D is d and angle between them is θ . If volume of tetrahedron is $\frac{1}{k} abd \sin \theta$ then k is
121. The perpendicular distance of a corner of unit cube from a diagonal not passing through it is k then $3k^2 =$
122. A square ABCD of diagonal 2a is folded along the diagonal AC so that planes DAC, BAC are at right angles. If the shortest distance between DC and AB is $\frac{ka}{\sqrt{3}}$ then k is

123. If $\vec{a}$ is a unit vector and projection of $\vec{x}$ along $\vec{a}$ is 2 units and $\vec{a} \times \vec{x} + \vec{b} = \vec{x}$ and $\vec{x} = \frac{(l\vec{a} + \vec{b} + \vec{a} \times \vec{b})}{m}$, then $l + m$ is

124. If acute angle between the line $\vec{r} = \vec{i} + 2\vec{j} + l(4\vec{i} - 3\vec{k})$ and xy -plane is α and acute angle between the planes $x + 2y = 0$, $2x + y = 0$ is β then $\cos^2 \alpha + \sin^2 \beta =$

KEY-LEVEL-VI**SINGLE ANSWER QUESTIONS**

1.B	2.B	3.B	4.D
5.B	6.A	7.B	8.B
9.A	10.A	11.B	12.D
13.A	14.C	15.D	16.C
17.A	18.A	19.B	20.A
21.A	22.A	23.C	24.A
25.B	26.B	27.B	28.A
29.C	30.C	31.C	32.D
33.C	34.A	35.B	36.B
37.B	38.A	39.B	40.B
41.A	42.B	43.A	44.C
45.B	46.A	47.A	48.C

MULTIANSWER QUESTIONS

49.ABCD	50.AC	51.AD	52.ABCD
53.AB	54.ABC	55.AB	56.ABD
57.BC	58.BC	59.AC	60.AB
61.ABC	62.BC	63.AB	64.ABCD
65.ABC	66.BD	67.AD	68.BD
69.BD	70.AC	71.BC	

ASSERTION-REASONING QUESTIONS

72. C	73. A	74. D	75. A
76. A	77. A	78. A	79. D
80. A	81. C	82. B	83. A
84. A	85. D	86. A	

PASSAGE QUESTIONS

87. B	88. B	89. A	90. B
91. C	92. A	93. A	94. C
95. D	96. A	97. B	98. B

MATRIX MATCHING QUESTIONS

99. A-q, B-p, C-s, D-r
 100. A-r, B-s, C-p, D-q
 101. A-s, B-r, C-p, D-q

INTEGER QUESTIONS

102.3	103.0	104.5	105.6
106.1	107.6	108.5	109.5
110.3	111.3	112(i).2	112(ii).1
113.9	114.2	115.6	116.5
117.1	118.3.	119.8	120.6
121.2	122.2	123.4	124.1

HINTS - LEVEL-VI**SINGLE ANSWER QUESTIONS**

1. Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$, where $x^2 + y^2 + z^2 = 1$

Unit vector along $3\hat{i} + 4\hat{j}$ is $\frac{3\hat{i} + 4\hat{j}}{5}$

The bisector of these two is $-\hat{i} + \hat{j} - \hat{k}$ (given).

Therefore

$$-\hat{i} + \hat{j} - \hat{k} = \lambda \left(x\hat{i} + y\hat{j} + z\hat{k} + \frac{3\hat{i} + 4\hat{j}}{5} \right)$$

$$-\hat{i} + \hat{j} - \hat{k} = \frac{1}{5} \lambda \left[(5x + 3)\hat{i} + (5y + 4)\hat{j} + 5z\hat{k} \right]$$

$$\frac{\lambda}{5}(5x + 3) = -1, \frac{\lambda}{5}(5y + 4) = 1, \frac{\lambda}{5}5z = -1$$

$$x = -\frac{5 + 3\lambda}{5\lambda}, y = -\frac{5 - 4\lambda}{5\lambda}, z = -\frac{1}{\lambda}$$

putting these values in (ie), i.e.,

$$x^2 + y^2 + z^2 = 1, \text{ we get}$$

$$(5+3\lambda)^2 + (5-4\lambda)^2 + 25 = 25\lambda^2$$

$$\lambda = 15/2$$

$$\therefore \vec{c} = \frac{1}{15}(-11\hat{i} + 10\hat{j} - 2\hat{k})$$

2. The velocity of wind relative to man
= Actual velocity of wind - Actual velocity of man

Let $\hat{i}$ and $\hat{j}$ represent unit vectors along east and north. Let the actual velocity of wind be given by $x\hat{i} + y\hat{j}$.

In the first case the man's velocity is $8\hat{i}$ and that of the wind blowing from the north relative to the man is $-p\hat{j}$. Therefore

$$-p\hat{j} = (x\hat{i} + y\hat{j}) - 8\hat{i} \quad (\text{by (i)})$$

Comparing coefficients $x-8=0$, $y=-p$

In the second case when the man doubles his speed, it seems to come from the north-east

$$-q(\hat{i} + \hat{j}) = (x\hat{i} + y\hat{j}) - 16\hat{i}$$

$$\therefore x-16 = -q, y = -q$$

Putting $x=8$, we get $q=8$

$$y=-8$$

Hence, the velocity of wind is

$$x\hat{i} + y\hat{j} = 8(\hat{i} - \hat{j})$$

Its magnitude is

$$\sqrt{(8^2 + 8^2)} = 8\sqrt{2} \quad \text{and} \quad \tan \theta = -1$$

$\theta = -45^\circ$. Hence, its direction is from the north-west

$$3. \quad \vec{AL} = \frac{\vec{AB} + \vec{AC}}{2}, \quad \vec{AM} = \frac{\vec{AC} + \vec{AD}}{2}$$

$$\therefore \vec{AL} + \vec{AM} = \frac{3}{2}\vec{AC}$$

$$4. \quad \sqrt{3} \tan \theta + 1 = 0 \text{ and } \sqrt{3} \sec \theta - 2 = 0$$

$$\Rightarrow \theta = \frac{11\pi}{6} \Rightarrow 2n\pi + \frac{11\pi}{6}, n \in \mathbb{Z}$$

$$5. \quad \therefore \vec{b} = \frac{5\hat{i} + 5k\hat{j}}{k+1}$$

$$\text{Also } |\vec{b}| \leq \sqrt{37}$$

$$\Rightarrow \frac{1}{k+1} \sqrt{25 + 25k^2} \leq \sqrt{37}$$

$$\Rightarrow 5\sqrt{1+k^2} \leq \sqrt{37}(k+1)$$

squaring both sides

$$25(1+k^2) \leq 37(k^2 + 2k + 1)$$

$$\text{or } 6k^2 + 37k + 6 \geq 0 \Rightarrow (6k+1)(k+6) \geq 0$$

$$k \in (-\infty, -6) \cup \left[-\frac{1}{6}, \infty\right)$$

6. Let $\vec{p} = x\vec{i} + y\vec{j}$ lies on the ellipse

Now from the diagram, according to given conditions, $AP=AB$

$$\text{or } (x + \sqrt{3})^2 + (y-0)^2 = 4 \text{ or } (x + \sqrt{3})^2 + y^2 = 4$$

solving (i) and (ii) we get $x=0$ and $y = \pm 1$

Hence point P has position vector $\pm \hat{j}$

7. If a, b, c are sides of a triangle then length of median through the vertex c is

$$\frac{1}{2} \sqrt{2b^2 + 2a^2 - c^2}$$

8. Given that $\vec{OA}, \vec{OB}$ and $\vec{OC}$ are coplanar,

then $\vec{OB}$ can be written as the linear combination of two non-collinear vectors $\vec{OA}$ and $\vec{OC}$. Let

$\vec{OC}$ and $\vec{OA}$ be taken along x and y axes respectively, and $|\vec{OA}| = |\vec{OB}| = |\vec{OC}| = R$

(radius). Clearly, we can say

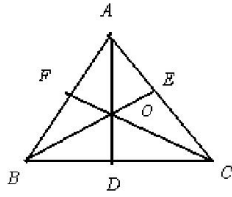
$$\vec{OB} = (\cos 60^\circ) \vec{OC} + (\cos 30^\circ) \vec{OA}$$

$$\vec{b} = \frac{1}{2} \vec{OC} + \frac{\sqrt{3}}{2} \vec{a}, \quad \vec{OC} = 2\vec{b} - \sqrt{3}\vec{a}$$

9. Given that A, B, C and O points are coplanar.

Let $\vec{a}, \vec{b}, \vec{c}$ and $\vec{w}$ be the position vectors of the points A, B, C, and O respectively. By the concept of coplanarity of 4 points, we have

$$x\vec{a} + y\vec{b} + z\vec{c} + t\vec{w} = \vec{0} \quad \text{and} \quad x + y + z + t = 0$$



$$\Rightarrow \frac{x\vec{a} + t\vec{w}}{x+t} = \frac{y\vec{b} + z\vec{c}}{y+z} = \text{Position vector of D point}$$

$$\Rightarrow \frac{OD}{AD} = \frac{x}{t} \dots\dots\dots (i)$$

$$\text{similarly, } \frac{OE}{BE} = \frac{y}{t} \dots\dots\dots (ii)$$

$$\frac{OF}{CF} = \frac{z}{t} \dots\dots\dots (iii)$$

From (i), (ii) & (iii), we get

$$\Rightarrow \left| \frac{OD}{AD} + \frac{OE}{BE} + \frac{OF}{CF} \right| = \left| \frac{x+y+z}{t} \right| = 1 = \text{e th}$$

10. Let the origin of reference be O, the circumcenter of the triangle and let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, $\vec{OC} = \vec{c}$ and $\vec{OT} = \vec{t}$

then $|\vec{a}| = |\vec{b}| = |\vec{c}| = R$ (circumradius)

Again

$$\vec{OA} + \vec{OB} + \vec{OC} = \vec{OA} + 2\vec{OD} = \vec{OA} + \vec{AH} = \vec{OH}$$

$\therefore$ P.V. of H is $\vec{a} + \vec{b} + \vec{c}$, since, D is the mid point of HT, we have

$$\frac{\vec{a} + \vec{b} + \vec{c} + \vec{t}}{2} = \frac{\vec{b} + \vec{c}}{2} \Rightarrow \vec{t} = -\vec{a}$$

$$\therefore \vec{AT} = -2\vec{a} \Rightarrow AT = |-2\vec{a}| = 2|\vec{a}| = 2R. \text{ But}$$

$$BC = 2R \sin A = R,$$

$$\therefore AT = 2BC$$

11. Let $\vec{OA} = \vec{b}$ and $\vec{OC} = \vec{c}$ then $\vec{AB} = \vec{b} - \vec{a}$ and

$$\vec{OP} = \frac{1}{3}\vec{a}, \vec{OQ} = \frac{1}{3}\vec{b}, \vec{OR} = \frac{1}{3}\vec{c}$$

since P, Q, R and S are coplanar, then

$\vec{PS} = \alpha\vec{PQ} + \beta\vec{PR}$ ($\vec{PS}$ can be written as a linear combination of $\vec{PQ}$ and $\vec{PR}$)

$$= \alpha(\vec{OQ} - \vec{OP}) + \beta(\vec{OR} - \vec{OP})$$

$$\text{i.e., } \vec{OS} - \vec{OP} = -(\alpha + \beta)\frac{\vec{a}}{3} + \frac{\alpha}{2}\vec{b} + \frac{\beta}{3}\vec{c}$$

$$\Rightarrow \vec{OS} = (1 - \alpha + \beta)\frac{\vec{a}}{3} + \frac{\alpha}{2}\vec{b} + \frac{\beta}{3}\vec{c} \dots\dots(1)$$

Given

$$\vec{OS} = \lambda\vec{AB} = \lambda(\vec{b} - \vec{a}) \dots\dots(2)$$

$$\text{From (1) and (2), } \beta = 0, \frac{1 - \alpha}{3} = -\lambda \text{ and}$$

$$\frac{\alpha}{2} = \lambda \Rightarrow 2\lambda + 3\lambda \Rightarrow \lambda = -1$$

12. Let p.v. of A, B, C be $\vec{a}, \vec{b}, \vec{c}$ respectively

Let $BC = \alpha, AC = \beta, AB = \gamma$

$$\text{p.v. of } pD = \frac{\beta\vec{b} + \gamma\vec{c}}{\beta + \gamma}$$

$$\text{p.v. of } D' = \frac{\beta\vec{b} - \gamma\vec{c}}{\beta - \gamma}$$

p.v. of mid $DD' =$ p.v. of

$$A' = \frac{\frac{\beta\vec{b} + \gamma\vec{c}}{\beta + \gamma} + \frac{\beta\vec{b} - \gamma\vec{c}}{\beta - \gamma}}{2}$$

$$= \frac{\beta^2\vec{b} - \gamma^2\vec{c}}{\beta^2 - \gamma^2} = \vec{a}' (\text{say})$$

$$\text{Similarly } \vec{b}' = \frac{\gamma^2\vec{c} - \alpha^2\vec{a}}{\gamma^2 - \alpha^2}$$

$$\vec{c}' = \frac{\alpha^2\vec{a} - \beta^2\vec{b}}{\alpha^2 - \beta^2}$$

Clearly

$$(\beta^2 - \alpha^2)\vec{a}' + (\gamma^2 - \alpha^2)\vec{b}' + (\alpha^2 - \beta^2)\vec{c}' = 0$$

$$\text{and } \beta^2 - \gamma^2 + \gamma^2 - \alpha^2 + \alpha^2 - \beta^2 = 0$$

$\therefore A', B', C'$ are collinear

13. Let A be the origin.

Let P.V. of $B = \vec{b}$, P.V. of $C = \vec{c}$, P.V. of $D = \vec{d}$

$$\text{P.V. of } P = \frac{\lambda_1\vec{b}}{\lambda_1 + 1} = \vec{p}$$

$$\text{P.V. of } Q = \frac{\lambda_2\vec{c} + \vec{b}}{\lambda_2 + 1} = \vec{q}$$

$$\text{P.V. of } R = \frac{\lambda_3\vec{d} + \vec{c}}{\lambda_3 + 1} = \vec{r} \quad \text{P.V. of } S = \frac{\vec{d}}{\lambda_4 + 1} = \vec{s}$$

$$(\lambda_2 + \lambda_1)\vec{q} = \lambda_2\vec{c} + \vec{b} \rightarrow (1)$$

$$(\lambda_3 + 1)\vec{r} = \lambda_3\vec{d} + \lambda_2\vec{c} \rightarrow (2)$$

(1) and (2)

$$\Rightarrow (\lambda_2 + 1)\vec{q} - (\lambda_3 + 1)\lambda_2\vec{r} = \vec{b} - \lambda_2\lambda_3\vec{d} \rightarrow (1)$$

$$\text{Now } \vec{b} = \left(\frac{\lambda_1 + 1}{\lambda_1} \right) \vec{p} \rightarrow (3)$$

$$\vec{d} = (\lambda_4 + 1)\vec{s}$$

$$\lambda_2\lambda_3\vec{d} = (\lambda_2\lambda_3)(\lambda_4 + 1)\vec{c} \rightarrow (4)$$

$$(3) \text{ and } (4) \quad \vec{b} - \lambda_2\lambda_3\vec{d}$$

$$= \left(\frac{\lambda_1 + 1}{\lambda_1} \right) \vec{p} - (\lambda_2\lambda_3)(\lambda_4 + 1)\vec{s} \rightarrow (2)$$

$$\text{From (1) and (2)} \quad (\lambda_1 + 1)\vec{q} - (\lambda_3 + 1)\lambda_2\vec{r}$$

$$= \left(\frac{\lambda_1 + 1}{\lambda_1} \right) \vec{p} - (\lambda_2\lambda_3)(\lambda_4 + 1)\vec{s}$$

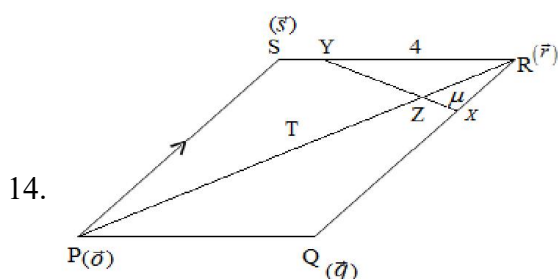
$$\therefore \left(\frac{\lambda_1 + 1}{\lambda_1} \right) \vec{p} - (\lambda_2 + 1)\vec{q} + (\lambda_3 + 1)\lambda_2\vec{r}$$

$$- \lambda_2\lambda_3(\lambda_4 + 1)\vec{s} = 0$$

similar P, Q, R, S are coplaner

$$\frac{\lambda_1 + 1}{\lambda_1} - (\lambda_2 + 1) + (\lambda_2 + 1)\lambda_2\lambda_3(\lambda_4 + 1) = 0$$

$$\Rightarrow \lambda_1\lambda_2\lambda_3\lambda_4 = 1$$



Let P be the origin

$$\text{Let } \vec{OQ} = \vec{q}, \vec{OR} = \vec{r}, \vec{OS} = \vec{s}$$

$$\vec{OR} = \vec{r} - \vec{q} = \vec{ps} = \vec{s}$$

$$\text{P.V. of } X = \frac{4\vec{r} + \vec{q}}{5}$$

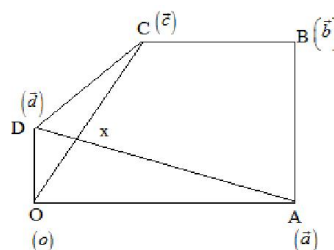
$$\text{P.V. of } Y = \frac{4\vec{s} + \vec{r}}{5}$$

$$= \frac{4(\vec{r} - \vec{q}) + \vec{r}}{5} = \vec{r} - \frac{4}{5}\vec{q}$$

$$\text{P.V. of } t = \frac{\mu \left(\vec{r} - \frac{4}{5}\vec{q} \right) + \frac{4\vec{r} + \vec{q}}{5}}{\mu + 1} = \frac{\lambda\vec{r}}{\lambda + 1}$$

$$\Rightarrow \frac{5\mu + 4}{5(\mu + 1)} = \frac{\lambda}{\lambda + 1} \text{ and } \frac{-4}{5}\mu + \frac{1}{5} = 0$$

$$\Rightarrow \mu = \frac{1}{4} \Rightarrow \frac{\lambda}{\lambda + 1} = \frac{\frac{25}{4}}{\frac{25}{4} + 1} \Rightarrow \lambda = \frac{25}{4}$$



Let OABCD be a pentagon in which the sides OA and CB are parallel and the sides OD and AB are parallel

$$\frac{OA}{CB} = \frac{2}{1} \text{ and } \frac{OD}{AB} = \frac{1}{2}$$

Let x be the point of intersection of OC and AD

$$\rightarrow \vec{OA} = 2\vec{CB} \Rightarrow \vec{a} = 2(\vec{b} - \vec{c})$$

$$3\vec{OD} = \vec{AB} \Rightarrow 3\vec{d} = \vec{b} - \vec{a}$$

$$= \vec{b} - 2\vec{b} + 2\vec{c}$$

$$= -\vec{b} + 2\vec{c}$$

$$\text{Let } \frac{OX}{XC} = \frac{\lambda}{1}, \frac{AX}{XD} = \frac{\mu}{1}$$

$$\Rightarrow \frac{\lambda\vec{c}}{\lambda + 1} = \frac{\mu}{\mu + 1} \left(\frac{-\vec{b} + 2\vec{c}}{3} \right) + \frac{2}{\mu + 1}(\vec{b} - \vec{c})$$

$$\Rightarrow \frac{-\mu}{3(\mu + 1)} + \frac{L}{\mu + 1} = 0 \Rightarrow \mu = 6$$

$$\Rightarrow \lambda = \frac{2}{5}$$

$$\therefore \frac{OX}{XC} = \frac{2}{5}$$

16. $f(x) = \frac{10-2x}{3(1-x)}$ and slop of tangent

drawn to $y = f(x)$ at $x = -1$ is $\frac{2}{3}$.

$$\text{Required vector} = m \left(\frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|} \right)$$

$$= \frac{2}{3\sqrt{6}}(\bar{i} + 2\bar{j} - \bar{k})$$

17. Let P be horizontal in the direction of unit vector $\hat{i}$. The resultant is also P but perpendicular to it in the direction of unit vector $\hat{j}$. If Q be the other force making an angle θ (obtuse) as resultant is perpendicular to P, then the two forces are $P\hat{i}$ and $Q\cos\theta\hat{i} + Q\sin\theta\hat{j}$ resultant is $P\hat{j}$.

$$\therefore P\hat{j} = P\hat{i} + (Q\cos\theta\hat{i} + Q\sin\theta\hat{j})$$

comparing coefficients of $\hat{i}$ and $\hat{j}$, we get

$$P + Q\cos\theta = 0 \text{ and } Q\sin\theta = P$$

$$\text{or } Q\cos\theta = -P \text{ and } Q\sin\theta = P$$

Squaring and adding $Q = P\sqrt{2}$ and dividing $\tan\theta = -1$

$$\theta = 135^\circ$$

18. $\vec{a} = (1, 3, \sin 2\alpha)$ makes an obtuse angle with the z-axis

$$\therefore \sin 2\alpha < 0$$

Since $\vec{b}$ and $\vec{c}$ are orthogonal

$$\Rightarrow \vec{b} \cdot \vec{c} = 0 \dots\dots\dots(1)$$

$$\therefore \tan^2 \alpha - \tan \alpha - 6 = 0 \Rightarrow \tan \alpha = 3 \text{ or } -2$$

If $\tan \alpha = 3$, then $\sin 2\alpha = \frac{2 \tan \alpha}{1 + \tan^2 \alpha} = \frac{3}{5} > 0$,

which is not possible (from (1))

$$\therefore \tan \alpha = -2.$$

$$\text{Again } \tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{4}{3} > 0.$$

$$\text{Also } \sin 2\alpha < 0$$

$\therefore 2\alpha$ lies in the third quadrant $\Rightarrow \frac{\alpha}{2}$ lies in the second quadrant

$$\therefore \sqrt{\sin \frac{\alpha}{2}} \text{ is valid and } \alpha = (4n+1)\pi - \tan^{-1} 2$$

19. Let image of P w.r.t. the given line be $Q(\alpha, \beta, \gamma)$

then mid point of PQ i.e. $\left(\frac{\alpha+7}{2}, \frac{\beta-1}{2}, \frac{\gamma+2}{2} \right)$

$$\text{lines on the line } \therefore \frac{\alpha+7}{2}\hat{i} + \frac{\beta-1}{2}\hat{j} + \frac{\gamma+2}{2}\hat{k} = 9\hat{i} + 5\hat{j} + 5\hat{k} + \lambda(\hat{i} + 3\hat{k} + 5\hat{k})$$

On comparing

$$\left. \begin{aligned} \frac{\alpha+7}{2} &= 9 + \lambda \Rightarrow \alpha = 11 + 2\lambda \\ \frac{\beta-1}{2} &= 5 + 3\lambda \Rightarrow \beta = 11 + 6\lambda \\ \frac{\gamma+2}{2} &= 5 + 5\lambda \Rightarrow \gamma = 8 + 10\lambda \end{aligned} \right\} \text{ and}$$

Also PQ and given line are $\perp$ i.e.,

$$(\alpha-7).1 + (\beta-1).3 + (\gamma-2).5 = 0$$

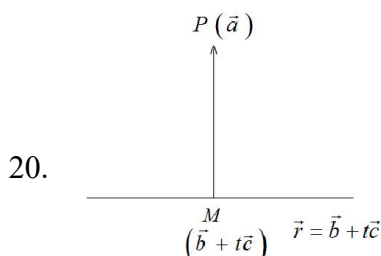
$$\Rightarrow \alpha - 7 + 3\beta + 3 + 5\gamma - 10 = 0$$

$$\therefore \alpha + 3\beta + 5\gamma = 14$$

From (1) and (2),

$$11 + 2\gamma + 3(11 + 6\lambda) + 5(8 + 10\lambda) = 14$$

$$\lambda = -1 \text{ From (1), } (\alpha, \beta, \gamma) = (9, 5, -2)$$



Let M be the foot of $\perp$ from P on to the line $\vec{r} = \vec{b} + t\vec{c}$

$$(\vec{a} - (\vec{b} + t\vec{c})) \cdot \vec{c} = 0$$

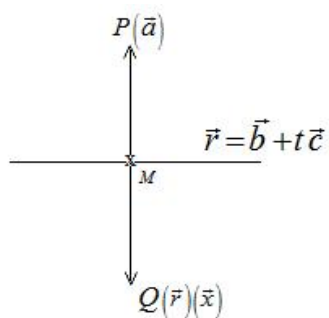
VECTOR ALGEBRA

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$$\vec{a} \cdot \vec{c} - \vec{b} \cdot \vec{c} - t |\vec{c}|^2 = 0$$

$$t = \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2}$$

$$\therefore \text{P.V of M} = \vec{b} + \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c}$$



21.

Let $Q(\vec{x})$ image pf $P(\vec{a})$ in

$$\vec{r} = \vec{b} + t \vec{c}$$

$$\text{Let } m = (\vec{b} + t \vec{c})$$

$$Pm \cdot \vec{c} = 0$$

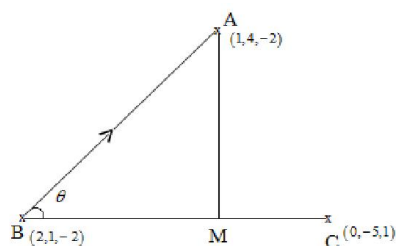
$$(\vec{b} + t \vec{c} - \vec{a}) \cdot \vec{c} = 0$$

$$\vec{b} \cdot \vec{c} + t |\vec{c}|^2 - \vec{a} \cdot \vec{c} = 0$$

$$t = \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2}$$

$$\text{Now } \frac{\vec{x} + \vec{a}}{2} = \vec{b} + \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c}$$

$$\vec{x} = 2 \left(\vec{b} + \frac{(\vec{a} - \vec{b}) \cdot \vec{c}}{|\vec{c}|^2} \vec{c} \right) - \vec{a}$$



22.

$$\text{Let } \vec{p} = \vec{BA}, \vec{q} = \vec{BC}$$

$$AM = BA \sin \theta = |\vec{p}| \sin \theta$$

$$|\vec{q} \times \vec{p}| = |\vec{q}| |\vec{p}| \sin \theta$$

$$AM = \frac{|\vec{q} \times \vec{p}|}{|\vec{q}|} = \frac{3\sqrt{26}}{7}$$

$$\begin{pmatrix} \vec{q} = -2\vec{i} - 6\vec{j} + 3\vec{k} \\ \vec{p} = -\vec{i} + 3\vec{j} \end{pmatrix}$$

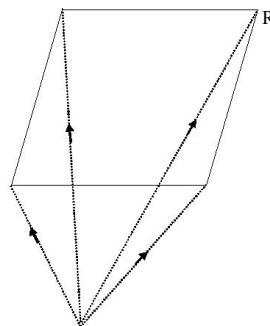
$$23. \quad |\vec{a} + 2\vec{b} + 3\vec{c}| = 3 + 2\sqrt{2}$$

$$4 \cos \alpha + 6 \cos \beta = -11 + 2\sqrt{2} - 12 \cos \theta$$

$$gqt \left[\frac{\pi}{2}, \frac{2\pi}{3} \right]$$

$$\text{max. of } 4 \cos \alpha + 6 \cos \beta \text{ is } 2\sqrt{2} - 5$$

$$24. \quad P: -2\hat{i} - \hat{j}, Q: 4\hat{i}, R: 3\hat{i} + 3\hat{j}, S: -3\hat{i} + 2\hat{j}$$



$$\overline{PQ} = \text{of } P = 6\hat{i} + \hat{j}$$

$$\overline{QR} = 3\hat{i} + 3\hat{j} - 4\hat{i} = -\hat{i} + 3\hat{j}$$

$$\overline{PS} = -3\hat{i} + 2\hat{j} + 2\hat{i} + \hat{j} = -\hat{i} + 3\hat{j}$$

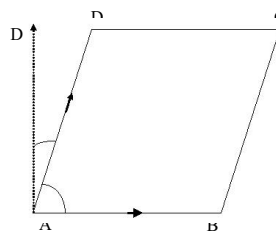
$$\overline{SR} = 3\hat{i} + 3\hat{j} + 3\hat{i} - 2\hat{j} = 6\hat{i} + \hat{j}$$

$$\overline{PQ} \cdot \overline{PS} = (6\hat{i} + \hat{j}) \cdot (-\hat{i} + 3\hat{j}) = -3 + 3 = 0$$

$$\text{Here } \overline{PQ} \parallel \overline{SR} \text{ and } \overline{PS} \parallel \overline{QR}$$

$$\text{but } \overline{PQ} \text{ is not perpendicular to } \overline{PS}$$

$$25. \quad \overline{AB} \cdot \overline{AD} > 0,$$



So θ is acute.

$$\cos q = \frac{\overrightarrow{AB} \cdot \overrightarrow{AD}}{|\overrightarrow{AB}| |\overrightarrow{AD}|} = \frac{8}{9}$$

$$\cos(a + q) = 0 \Rightarrow \cos a \cdot \cos q - \sin a \cdot \sin q = 0$$

$$8 \cos a - \sqrt{17} \cdot \sin a = 0$$

$$64 \cos^2 a = 17 \sin^2 a \Rightarrow \cos a = \frac{\sqrt{17}}{9}$$

26. We have: $\vec{A} + \vec{B} = \vec{a}$

$$\vec{A} \cdot \vec{a} + \vec{B} \cdot \vec{a} = \vec{a} \cdot \vec{a} \Rightarrow 1 + \vec{B} \cdot \vec{a} = a^2$$

$$(\text{given } \vec{A} \cdot \vec{a} = 1) \Rightarrow \vec{B} \cdot \vec{a} = a^2 - 1 \quad \dots\dots$$

(i)

$$\text{also } \vec{A} \times \vec{B} = \vec{b} \Rightarrow \vec{a} \times (\vec{A} \times \vec{B}) = \vec{a} \times \vec{b}$$

$$\Rightarrow (\vec{a} \cdot \vec{B}) \vec{A} - (\vec{a} \cdot \vec{A}) \vec{B} = \vec{a} \times \vec{b}$$

$$\Rightarrow (a^2 - 1) \vec{A} - \vec{B} = \vec{a} \times \vec{b}$$

$$(\text{using (i) and } \vec{A} \cdot \vec{a} = 1) \quad \dots\dots(ii)$$

$$\text{and } \vec{A} + \vec{B} = \vec{a}$$

$$\text{From (ii) and (iii), } \vec{A} = \frac{(\vec{a} \times \vec{b}) + \vec{a}}{a^2} \text{ and}$$

$$\vec{B} = \vec{a} - \left\{ \frac{(\vec{a} \times \vec{b}) + \vec{a}}{a^2} \right\} \text{ or}$$

$$\vec{B} = \frac{(\vec{b} \times \vec{a}) + \vec{a}(a^2 - 1)}{a^2}$$

$$\text{thus } \vec{A} = \frac{(\vec{a} \times \vec{b}) + \vec{a}}{a^2} \text{ and}$$

$$\vec{B} = \frac{(\vec{b} \times \vec{a}) + \vec{a}(a^2 - 1)}{a^2}$$

27. $\vec{r} \times \vec{a} = \vec{b} \times \vec{a} \rightarrow (1)$

$$\vec{r} \times \vec{b} = \vec{a} \times \vec{b} \rightarrow (2)$$

$$\vec{r} \times \vec{a} = -(\vec{r} \times \vec{b})$$

$$\vec{r} \times \vec{a} + \vec{r} \times \vec{b} = 0$$

$$\Rightarrow \vec{r} \times (\vec{a} + \vec{b}) = 0 \Rightarrow \vec{r} \parallel \vec{a} + \vec{b}$$

$$\therefore \vec{r} = t(\vec{a} + \vec{b})$$

$$\text{From (1) } t(\vec{a} + \vec{b}) \times \vec{a} = \vec{b} \times \vec{a}$$

$$t(\vec{b} \times \vec{a}) - (\vec{b} \times \vec{a})$$

$$\Rightarrow (\vec{b} \times \vec{a})(t - 1) = 0 \Rightarrow t = 1$$

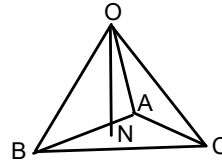
$$\therefore \vec{r} = \vec{a} + \vec{b}$$

28. normal vector of ABC is

$$(\overrightarrow{AB} \times \overrightarrow{AC})$$

$$\vec{b} - \vec{a} \times \vec{c} - \vec{a}$$

$$\vec{n} = \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}$$



Then $\perp$ distance from O to plane ABC is

$$\frac{\vec{a} \cdot \vec{n}}{|\vec{n}|} = \frac{|\vec{a} \cdot \vec{b} \times \vec{c}|}{|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|}$$

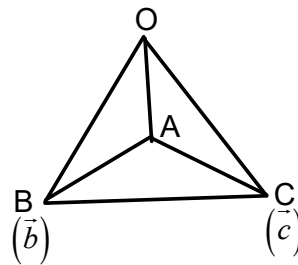
$$= \frac{[\vec{a} \vec{b} \vec{c}]}{2 \times \text{Area of } \Delta ABC}$$

$$= \sqrt{\frac{2}{3}} a$$

29. Consider $\overrightarrow{OB} = \vec{k}$

$$\text{its opposite edge } \overrightarrow{AC} = \vec{c} - \vec{a}$$

we have to find shortest distance between $\overrightarrow{OA}$ and $\overrightarrow{AC}$



$$\text{equation } \overrightarrow{OA} \text{ is } \vec{r} = 1 - \vec{b}$$

$$\text{equation } \overrightarrow{AC} \text{ is } \vec{r} = (1 - s)\vec{a} + s\vec{c}$$

$$\vec{r} = \vec{a} + s(\vec{c} - \vec{a})$$

S.D projection $\vec{a}$ on $\vec{b} \times (\vec{c} - \vec{a})$

$$\begin{aligned} &= \frac{\vec{a} \times (\vec{b} \times \vec{c} - \vec{b} \times \vec{a})}{|\vec{b} \times \vec{c} - \vec{b} \times \vec{a}|} = \frac{[\vec{a} \vec{b} \vec{c}]}{\vec{b} \times (\vec{c} - \vec{a})} \\ &= \frac{k^3}{\sqrt{2}} \sin 90^\circ \\ &= \frac{k^3}{\sqrt{2}k^2} = \left(\frac{k}{\sqrt{2}} \right) \end{aligned}$$

30. Using the formula

$$S.D = \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\vec{a}_2 - \vec{a}_1 = 4\vec{i} - 16\vec{j}$$

$$\vec{b}_1 \times \vec{b}_2 = 16\vec{i} + 16\vec{j} + 16\vec{k}$$

$$\frac{\vec{b}_1 \times \vec{b}_2}{|\vec{b}_1 \times \vec{b}_2|} = \frac{1}{\sqrt{3}}(\vec{i} + \vec{j} + \vec{k})$$

$$\text{Hence S.D} = \frac{1}{\sqrt{3}}|4 - 16| = \frac{12}{\sqrt{3}} = 4\sqrt{3}$$

31. Vector perpendicular to plane which is parallel to vectors $i + j + k$ and $2i$

$$\vec{a} = (i + j + k) \times (2i) = 2(j - k)$$

Vector perpendicular to plane which is parallel to vectors $i + j$ and $i - k$

$$\vec{b} = (i + j) \times (i - k) = -i + j - k$$

Vector parallel to the line of intersection of the planes $\hat{a} \times \hat{b} = 2j + 2k$

Angle of intersection between $2i - j$ and the line of intersection of the planes

$$= \cos^{-1} \left(\frac{(\vec{a} \times \vec{b}) \cdot (2i - j)}{|\vec{a} \times \vec{b}| |2i - j|} \right)$$

$$\begin{aligned} &= \cos^{-1} \left(\frac{(2j + 2k) \cdot (2i - j)}{\sqrt{4 + 4} \sqrt{4 + 1}} \right) \\ &= \cos^{-1} \left(\frac{-2}{\sqrt{8} \sqrt{5}} \right) = \cos^{-1} \left(-\frac{1}{10} \right) \\ &= \pi - \cos^{-1} \left(\frac{1}{\sqrt{10}} \right) \end{aligned}$$

$$\text{Acute angle} = \cos^{-1} \left(\frac{1}{\sqrt{10}} \right)$$

$$\begin{aligned} 32. \quad &\vec{x} \times \vec{a} + (\vec{x} \cdot \vec{b})\vec{c} = \vec{d} \\ &\therefore \{ \vec{x} \times \vec{a} + (\vec{x} \cdot \vec{b})\vec{c} \} \times \vec{c} = \vec{d} \times \vec{c} \text{ or} \\ &(\vec{x} \cdot \vec{b})(\vec{c} \times \vec{c}) = \vec{d} \times \vec{c} \end{aligned}$$

$$\Rightarrow (\vec{x} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{x} = (\vec{d} \times \vec{c})$$

$$\Rightarrow a \times \{ (\vec{x} \cdot \vec{c})\vec{a} - (\vec{a} \cdot \vec{c})\vec{x} \} = \vec{a} \times (\vec{d} \times \vec{c})$$

$$\Rightarrow -(\vec{a} \cdot \vec{c})(\vec{a} \times \vec{x}) = \vec{a} \times (\vec{d} \times \vec{c})$$

$$(\therefore \vec{a} \times \vec{a} = 0)$$

$$\Rightarrow \vec{x} \times \vec{a} = \frac{\vec{a} \times (\vec{d} \times \vec{c})}{\vec{a} \cdot \vec{c}}$$

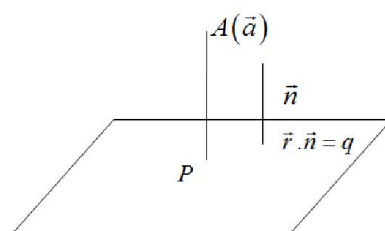
$$\Rightarrow \vec{a} \times (\vec{x} \times \vec{a}) = \vec{a} \times \frac{\vec{a} \times (\vec{d} \times \vec{c})}{\vec{a} \cdot \vec{c}}$$

$$(\vec{a} \cdot \vec{a})\vec{x} - (\vec{a} \cdot \vec{x})\vec{a} = \vec{a} \times \frac{\vec{a} \times (\vec{d} \times \vec{c})}{\vec{a} \cdot \vec{c}} \text{ or}$$

$$\vec{a} \cdot \vec{x} = (\vec{a} \cdot \vec{x})\vec{a} + \vec{a} \times \frac{\vec{a} \times (\vec{d} \times \vec{c})}{\vec{a} \cdot \vec{c}}$$

$$\Rightarrow \vec{x} = \frac{(\vec{a} \cdot \vec{x})\vec{a}}{\vec{a}^2} + \vec{a} \times \frac{\vec{a} \times (\vec{d} \times \vec{c})}{(\vec{a} \cdot \vec{c})\vec{a}^2}$$

$$= \lambda \vec{a} + \vec{a} \times \frac{\vec{a} \times (\vec{d} \times \vec{c})}{(\vec{a} \cdot \vec{c})\vec{a}^2}; \text{ Where } \lambda = \frac{\vec{a} \cdot \vec{x}}{\vec{a}^2}$$



33.

Any point p on the line passing through $A(\vec{a})$ and parallel to $\vec{n}$ is

$$\vec{r} = \vec{a} + t\vec{n}$$

This point AP lines on $\vec{r} \cdot \vec{n} = q$

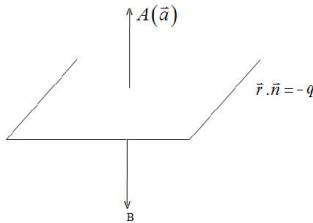
$$(\vec{a} + t\vec{n}) \cdot \vec{n} = q$$

$$\vec{a} \cdot \vec{n} + t|\vec{n}|^2 = q$$

$$t = \frac{q - \vec{a} \cdot \vec{n}}{|\vec{n}|^2}$$

$$\begin{aligned} \therefore |\vec{AP}| &= |\vec{OP} - \vec{OA}| = |\vec{v} - \vec{a}| \\ &= \left| \frac{\vec{a} + t\vec{n} - \vec{a}}{|\vec{n}|^2} \right| |\vec{n}| = \left| \frac{\vec{a} - \vec{a} \cdot \vec{n}}{|\vec{n}|} \right| \end{aligned}$$

34.



Let B be the reflecting A in the plane $\vec{r} \cdot \vec{n} = q$ and A line $\vec{AB}$ is

$$\vec{r} = \vec{a} + t\vec{n}$$

Let P.V. (B) = $(\vec{a} + t\vec{n})$

Mid(B) = $\frac{\vec{a} + \vec{a} + t\vec{n}}{2}$ lies in the plane = $\vec{a} + \frac{t}{2}\vec{n}$

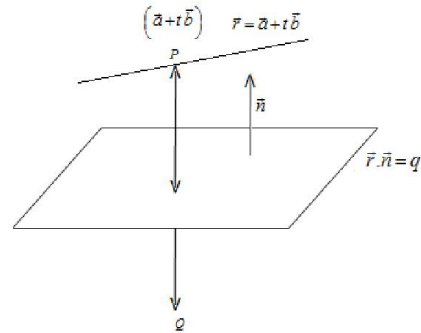
$$\left(\vec{a} + \frac{t}{2}\vec{n} \right) \cdot \vec{n} = q$$

$$\vec{a} \cdot \vec{n} + \frac{t}{2}|\vec{n}|^2 = q$$

$$t = \frac{2(q - \vec{a} \cdot \vec{n})}{|\vec{n}|^2}$$

$$\therefore B = \vec{a} + \frac{2(\vec{a} - \vec{a} \cdot \vec{n})}{|\vec{n}|^2}$$

35.



Let $P(\vec{a} + t\vec{b})$ be any point on the line

Equation of the line $\vec{PQ}$ is

$$\vec{r} = (\vec{a} + t\vec{b}) + \lambda \vec{n}$$

Let P.V. of Q is

$$\vec{a} + t\vec{b} + \lambda \vec{n}$$

$$\text{Mid}(PQ) = \left(\frac{\vec{a} + t\vec{b} + \vec{a} + t\vec{b} + \lambda \vec{n}}{2} \right)$$

$$= \vec{a} + t\vec{b} + \frac{\lambda}{2}\vec{n} \text{ lies on } \vec{r} \cdot \vec{n} = q$$

$$\left(\vec{a} + t\vec{b} + \frac{\lambda}{2}\vec{n} \right) \cdot \vec{n} = q$$

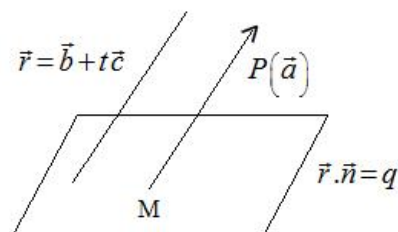
$$\vec{a} \cdot \vec{n} + t\vec{b} \cdot \vec{n} + \frac{\lambda}{2}|\vec{n}|^2 = q$$

$$\lambda = \frac{2(q - (\vec{a} + t\vec{b}) \cdot \vec{n})}{|\vec{n}|}$$

$$\therefore \text{P.V of } Q = \vec{a} + t\vec{b} + \frac{2(q - (\vec{a} + t\vec{b}) \cdot \vec{n})}{|\vec{n}|} \vec{n}$$

$$\vec{a} + \frac{2(q - \vec{a} \cdot \vec{n})}{|\vec{n}|^2} \vec{n} + t \left(\vec{b} - 2 \frac{(\vec{b} \cdot \vec{n})}{|\vec{n}|^2} \vec{n} \right)$$

36.



VECTOR ALGEBRA

JEE ADVANCED - VOL - I

Any point on the line $\overrightarrow{PM}$ is

$$\vec{r} = \vec{a} + t\vec{c}$$

Suppose it represents M

$M(\vec{a} + t\vec{c})$ lies on $\vec{r} \cdot \vec{n} = q$

$$(\vec{a} + t\vec{c}) \cdot \vec{n} = q$$

$$\vec{a} \cdot \vec{n} + t(\vec{c} \cdot \vec{n}) = q$$

$$t = \frac{q - \vec{a} \cdot \vec{n}}{\vec{c} \cdot \vec{n}}$$

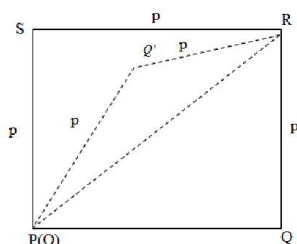
$$|\overrightarrow{MP}| = |\vec{a} - (\vec{a} + t\vec{c})|$$

$$= |t\vec{c}|$$

$$= |t| |\vec{c}|$$

$$= \left| \frac{(q - \vec{a} \cdot \vec{n})}{\vec{c} \cdot \vec{n}} \right| |\vec{c}|$$

37.



Let new positing Q be Q' . Let $\vec{i}, \vec{j}, \vec{k}$ denote unit vector along PQ' .

Let $\overrightarrow{OP} = \vec{0}$

P.V. of Q' be $p\vec{i}$

P.V. of R be $p\vec{i} + p\vec{j}$

P.V. of S be $p\vec{k}$

Eq of $\overrightarrow{PQ}$

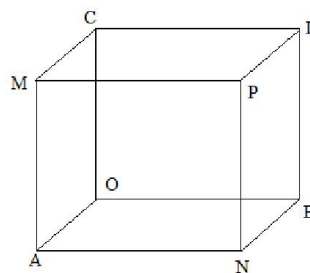
$$\vec{r} = t p \vec{i}$$

Eq of RS $\vec{r} = p\vec{i} + p\vec{j} + t(p\vec{i} + p\vec{j} - p\vec{k})$.

required shortest distance

$$= \frac{|(p\vec{i} + p\vec{j}) \cdot p\vec{i} \times (p\vec{i} + p\vec{j} - p\vec{k})|}{|p\vec{i} + (p\vec{i} + p\vec{j} - p\vec{k})|} = \frac{p}{\sqrt{2}}$$

38.



Let $\overrightarrow{OA} = x\vec{i}, \overrightarrow{OB} = y\vec{j}, \overrightarrow{OC} = z\vec{k}$ where

$\vec{i}, \vec{j}, \vec{k}$ are unit vectors along OA, OB, OC

diagonal CN and BM are

skew to their edge $\overrightarrow{OA}$.

P.V. of A $\rightarrow x\vec{i}$

P.V. of C $\rightarrow z\vec{k}$

P.V. of N $\rightarrow x\vec{i} + y\vec{j}$

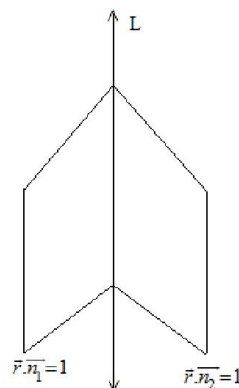
Eq of $\overrightarrow{OA}$ is $\vec{r} = t\vec{i}$

P.V. of CN is $\vec{r} = z\vec{k} + p(x\vec{i} + y\vec{j} - z\vec{k})$

Non shortest Distance between OA and CN

$$= \frac{|(z\vec{k} - \vec{0}) \cdot \vec{i} \times (x\vec{i} + y\vec{j} - z\vec{k})|}{|\vec{i} \times (x\vec{i} + y\vec{j} - z\vec{k})|} = \frac{yz}{\sqrt{y^2 + z^2}}$$

39.



Clearly $\vec{n}_1 \times \vec{n}_2 \parallel$ the line L

(Inter putting two planes)

required line passing through $\vec{a}$

and $\parallel$ to $\vec{n}_1 \times \vec{n}_2$

$\therefore$ its & is

$$\vec{v} = \vec{a} + t(\vec{n}_1 \times \vec{n}_2)$$

40. Clearly $\vec{n}_1 \times \vec{n}_2$ is parallel is line of intersation of

$\vec{r} \cdot \vec{n}_1 = q_1$ and $\vec{r} \cdot \vec{n}_2 = q_2$.

required plane $\vec{r} \cdot \vec{n}_2 = q_2 \perp$ of the plane

$\Rightarrow \vec{n}_1 \times \vec{n}_2 \perp^{\text{les}}$ required plane

required plane is how $\vec{n}_1 \times \vec{n}_2$ as its n or me

$\therefore$ its equation is $\vec{r} \cdot \vec{n}_1 \times \vec{n}_2 = \lambda$

its passing through $\vec{a}$

$\therefore \lambda = \vec{a} \cdot \vec{n}_1 \times \vec{n}_2$

$\therefore$ Equation plane is $\vec{r} \cdot \vec{n}_1 \times \vec{n}_2 = \vec{a} \cdot \vec{n}_1 \times \vec{n}_2$

41. $\vec{a} \times \vec{b} = x\vec{p} + y\vec{q} + z\vec{r}$

$$(\vec{a} \times \vec{b}) \cdot (\vec{q} \times \vec{r}) = x[\vec{p} \cdot \vec{q} \times \vec{r}]$$

$$x = \frac{1}{[\vec{p} \cdot \vec{q} \times \vec{r}]} (\vec{a} \times \vec{b} \cdot \vec{q} \times \vec{r})$$

$$= \frac{1}{[\vec{p} \cdot \vec{q} \times \vec{r}]} \begin{vmatrix} \vec{a} \cdot \vec{q} & \vec{a} \cdot \vec{r} \\ \vec{b} \cdot \vec{q} & \vec{b} \cdot \vec{r} \end{vmatrix}$$

$$y = \frac{1}{[\vec{p} \cdot \vec{q} \times \vec{r}]} \begin{vmatrix} \vec{a} \cdot \vec{p} & \vec{b} \cdot \vec{q} \\ \vec{a} \cdot \vec{r} & \vec{b} \cdot \vec{r} \end{vmatrix}$$

$$z = \frac{1}{[\vec{p} \cdot \vec{q} \times \vec{r}]} \begin{vmatrix} \vec{a} \cdot \vec{p} & \vec{b} \cdot \vec{p} \\ \vec{a} \cdot \vec{q} & \vec{b} \cdot \vec{q} \end{vmatrix}$$

$$\therefore \vec{a} \times \vec{b}$$

$$= \frac{1}{\sqrt{[\vec{p} \times \vec{q}, \vec{q} \times \vec{r}, \vec{r} \times \vec{p}]}}$$

$$\begin{vmatrix} \vec{p} & \vec{q} & \vec{r} \\ \vec{p} \cdot \vec{q} & \vec{q} \cdot \vec{a} & \vec{r} \cdot \vec{a} \\ \vec{p} \cdot \vec{b} & \vec{q} \cdot \vec{b} & \vec{r} \cdot \vec{b} \end{vmatrix}$$

42. Equation of plane passing through intersecting two planes

$\vec{r} \cdot \vec{n}_1 + \vec{n}_2 + \lambda \vec{n}_2 = p_1 + \lambda p_2$. Where λ is a parameter

This plane and $\vec{r} \cdot \vec{n}_3 = p_3$ are identical for same value of λ .

$$\vec{n}_1 + \lambda \vec{n}_2 = k \vec{n}_3$$

$$p_1 + \lambda p_2 = k p_3$$

$$\Rightarrow \vec{n}_1 \times \vec{n}_3 + \lambda \vec{n}_2 \times \vec{n}_3 = 0$$

$$\vec{n}_1 \times \vec{n}_2 = k \vec{n}_3 \times \vec{n}_2$$

$$\therefore (p_1 + \lambda p_2)(\vec{n}_2 \times \vec{n}_3) = k p_3 (\vec{n}_2 \times \vec{n}_3) = p \vec{n}_2 \times \vec{n}_1$$

$$\therefore (\vec{n}_2 \times \vec{n}_3) + p_2 \lambda (\vec{n}_2 \times \vec{n}_3) + p_3 \vec{n}_1 \times \vec{n}_2 = 0$$

$$\Rightarrow p_1 (\vec{n}_2 \times \vec{n}_3) + p_2 (\vec{n}_3 \times \vec{n}_1) + p_3 (\vec{n}_1 \times \vec{n}_2) = 0$$

43.
$$v = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= |a_1 b_1 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_1 b_3 c_2 - a_3 b_2 c_1|$$

$$AM \geq GM$$

$$\frac{a_1 + b_2 + c_3}{3} \geq (a_1 b_2 c_3)^{1/3}$$

$$\Rightarrow a_1 b_2 c_3 \leq \left(\frac{a_1 + b_2 + c_3}{3} \right)^3$$

$$a_2 b_3 c_1 \leq \left(\frac{a_2 + b_3 + c_1}{3} \right)^3$$

$$a_3 b_1 c_2 \leq \left(\frac{a_3 + b_1 + c_2}{3} \right)^3$$

$$v = a_1 b_2 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2$$

$$\leq \frac{1}{27} \left((a_1 + b_2 + c_3)^3 + (a_2 + b_3 + c_1)^3 + (a_3 + b_1 + c_2)^3 \right)$$

$$\leq \frac{1}{27} (a_1 + b_2 + c_3 + a_2 + b_3 + c_1 + a_3 + b_1 + c_2)^3$$

$$= \frac{1}{27} (3c)^2 = c^3$$

44. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$ and

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$$

$$[\vec{a} \vec{b} \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix} \text{ Let. } \cos \theta = m$$

$$\begin{aligned}
 &= 1(1 - m^2) - m(m - m^2) + m(m^2 - m) \\
 &= 1 - m^2 - m^2 + m^3 + m^3 - m^2 \\
 &= 2m^3 - 3m^2 + 1 \geq 0 \\
 &= (m - 1)^2(2m + 1) \geq 0 \\
 &m \geq \frac{-1}{2} \Rightarrow 0 \leq \theta \leq \frac{2\pi}{3} \\
 &\Rightarrow \text{Max. value of } \theta \text{ is } \frac{2\pi}{3}
 \end{aligned}$$

45. (B) We observe that

$$\vec{a} \cdot \vec{b}_1 = \vec{a} \cdot \vec{b} - \left(\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|^2} \right) \vec{a} \cdot \vec{a} = \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{b} = 0$$

$$\vec{a} \cdot \vec{c}_2 = \vec{a} \cdot \left(\frac{\vec{c} \cdot \vec{a}}{|\vec{a}|^2} \vec{a} - \frac{\vec{c} \cdot \vec{b}_1}{|\vec{b}_1|^2} \vec{b}_1 \right)$$

$$= \vec{a} \cdot \vec{c} - \vec{a} \cdot \vec{c} - 0 = 0 \quad [\because \vec{a} \cdot \vec{b}_1 = 0]$$

46. Let volume is V

$$V^2 = [\hat{a} \hat{b} \hat{c}]^2$$

$$= \begin{vmatrix} \hat{a} \cdot \hat{a} & \hat{a} \cdot \hat{b} & \hat{a} \cdot \hat{c} \\ \hat{b} \cdot \hat{a} & \hat{b} \cdot \hat{b} & \hat{b} \cdot \hat{c} \\ \hat{c} \cdot \hat{a} & \hat{c} \cdot \hat{b} & \hat{c} \cdot \hat{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{vmatrix}$$

$$= \frac{1}{2}$$

47. $\vec{r} = \alpha \vec{a} + \beta \vec{b} + \gamma (\vec{a} \times \vec{b})$

$$\left. \begin{aligned} \vec{r} \cdot \vec{a} = 2 &\Rightarrow 4\alpha + \beta = 2 \\ \vec{r} \cdot \vec{b} = 8 &\Rightarrow \alpha + \beta = 8 \end{aligned} \right\} \Rightarrow \begin{aligned} \alpha &= -2 \\ \beta &= 10 \end{aligned}$$

$$(\vec{r} + 2\vec{a} - 10\vec{b}) = \lambda (\vec{a} \times \vec{b})$$

$$(\vec{r} + 2\vec{a} - 10\vec{b}) \cdot (\vec{a} \times \vec{b}) = \lambda |\vec{a} \times \vec{b}|^2$$

$$\Rightarrow \lambda = \frac{4}{\sqrt{3}}$$

48. $\vec{v} = \lambda \vec{a} + \mu \vec{b}$

$$= \lambda (\hat{i} + \hat{j} + \hat{k}) + \mu (\hat{i} - \hat{j} - \hat{k})$$

Projection of $\vec{v}$ on $\vec{c}$ is

$$\frac{\vec{v} \cdot \vec{c}}{|\vec{c}|} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \frac{[(\lambda + \mu)\hat{i} + (\lambda - \mu)\hat{j} + (\lambda + \mu)\hat{k}] \cdot (\hat{i} - \hat{j} - \hat{k})}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{3}}$$

$$\Rightarrow \lambda + \mu - \lambda + \mu - \lambda - \mu = 1$$

$$\Rightarrow \mu - \lambda = 1$$

$$\Rightarrow \lambda = \mu - 1$$

$$\Rightarrow \vec{v} = (\mu - 1)(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - \hat{j} - \hat{k}) = \mu$$

$$(2\hat{i} + 2\hat{k}) - \hat{i} - \hat{j} - \hat{k}$$

$$\Rightarrow \vec{v} = (2\mu - 1)\hat{i} - \hat{j} + (2\mu - 1)\hat{k}$$

$$\text{At } \mu = 2, \vec{v} = 3\hat{i} - \hat{j} + 3\hat{k}$$